

Degree Project in Technology

Second cycle, 30 credits

Modeling Psychotic Risk from Adolescent Substance Use

Silpa Soni Nallacheruvu

Master's Programme in Biostatistics and Data Science, 120 credits

Date: May 2026

Supervisors: Shuyang Yao, Sam Andersson

Examiner: Tom Britton

Karolinska Institutet

Abstract

Schizophrenia is widely understood as a multi-factorial disorder, yet how genetic, environmental, and behavioral factors interact to shape risk remains unclear. This study investigates how adolescent substance use interacts with these domains to influence positive and negative psychotic symptoms, using both structured statistical models (GLMM, GAMM) and a deep learning approach (DCMFNet).

Across models, interactions involving substance use emerge as central to risk, particularly in combination with adverse childhood experiences and genetic liability, including cell-type specific polygenic risk scores. These effects are not uniform but context-dependent, indicating that substance use acts as an amplifier rather than an independent predictor.

A key distinction is observed between symptom dimensions. Positive symptoms are characterized by stronger and more targeted interaction effects and benefit from more flexible modeling approaches. In contrast, negative symptoms exhibit weaker and more stable patterns, with limited gains from increased model complexity.

From a methodological perspective, DCMFNet extends traditional models by providing a more flexible representation of interaction structure. While maintaining comparable interpretability, it improves predictive performance and stability, particularly in identifying individuals at higher risk.

Overall, the findings suggest that schizophrenia risk is best understood not through individual factors, but through how genetic, environmental, and behavioral influences combine and evolve across developmental. This highlights the importance of modeling interaction structure in psychiatric research.

Keywords: Schizophrenia, Substance Use, Gene–Environment Interaction, Cell-Type Specific Polygenic Risk Score, Psychotic Symptoms, Machine Learning, Deep Learning, Interaction Modeling

Sammanfattning

Schizofreni är en multifaktoriell sjukdom där genetiska, miljömässiga och beteendemässiga faktorer bidrar till risken, men hur dessa samverkar är fortfarande inte fullt klarlagt. Denna studie syftar till att undersöka hur substansbruk i tonåren interagerar med dessa faktorer och påverkar positiva och negativa psykotiska symtom, med hjälp av både statistiska modeller (GLMM, GAMM) och en djupinlärningsmodell (DCMFNet).

Resultaten visar att interaktioner som involverar substansbruk är centrala för risk, särskilt i kombination med negativa barndomsupplevelser och genetisk sårbarhet, inklusive celltypsspecifika polygena riskscore. Effekterna är kontextberoende och varierar mellan olika faktorer, vilket tyder på att substansbruk fungerar som en förstärkande snarare än en självständig riskfaktor.

En tydlig skillnad observeras mellan symtomdimensionerna. Positiva symtom uppvisar starkare och mer riktade interaktionseffekter och gynnas av mer flexibla modelleringsmetoder. Negativa symtom uppvisar däremot svagare och mer stabila mönster, med begränsad förbättring vid ökad modellkomplexitet.

DCMFNet utgör ett komplement till traditionella modeller genom att möjliggöra en mer flexibel representation av interaktionsstrukturer. Modellen uppvisar förbättrad prediktionsförmåga och stabilitet, särskilt vid identifiering av individer med högre risk, samtidigt som en viss grad av tolkbarhet bibehålls.

Sammanfattningsvis tyder resultaten på att risken för schizofreni bäst förstås genom hur genetiska, miljömässiga och beteendemässiga faktorer samverkar. Detta understryker vikten av att beakta interaktioner vid modellering av psykiatriska utfall.

Nyckelord: Schizofreni, Substansbruk, Gen–miljöinteraktion, Celltypsspecifika Polygena riskscore, Psykotiska symtom, Maskininlärning, Djupinlärning, Interaktionsmodellering

Acknowledgments

I would like to sincerely thank my supervisors, Dr. Shuyang Yao and Sam Andersson, for their guidance, and for giving me the opportunity to work on this project. I am also grateful to Dr. Lu Yi and her team at MEB, with whom I had the pleasure of working closely, for their thoughtful insights and for bringing fresh perspectives that continually shaped this work. I would like to express my heartfelt appreciation to my friends for their companionship, encouragement throughout this journey. Finally, I am deeply grateful to my family, and to Shree Garg, for their unwavering support, believing in me, and for being a constant source of strength at every step of this journey.

Contents

Abstract	i
Sammanfattning	ii
Acknowledgments	iii
List of Abbreviations	vi
1 Introduction	1
1.1 Background and Motivation	1
1.2 Research Gap	2
1.3 Research Aim and Questions	3
2 Theory	4
2.1 Genetic Risk and Cell-type Polygenic Risk Scores	4
2.2 Statistical Modeling Approaches	4
2.3 Machine Learning and Deep Learning Approaches	5
3 Data	7
3.1 Study Cohort	7
3.2 Genotype Data and Cell-type PRS Generation	7
3.2.1 Cell-type PRS Pipeline	7
3.3 Phenotype Data	8
3.4 Feature Selection	10
3.5 Descriptive Statistics	12
3.6 Handling Missing Data	13
4 Methodology	14
4.1 Modeling Strategy	14
4.2 Statistical Models	14
4.2.1 Generalized Linear Mixed Model (GLMM)	14
4.2.2 Generalized Additive Mixed Model (GAMM)	15
4.3 Deep Learning Model	15
4.3.1 DCMFNet Architecture	15

4.3.2	Model Adaptation	17
4.4	Model Training and Evaluation	18
5	Results	19
5.1	GLMM	19
5.2	GAMM	20
5.3	DCMFNet	22
5.3.1	Default Model Training	22
5.3.2	Hyperparameter Tuning	23
5.4	Model Comparison	25
6	Analysis and Interpretation	28
6.1	GLMM	28
6.2	GAMM	30
6.3	DCMFNet	32
6.3.1	Complexity Analysis	32
6.3.2	Importance Analysis	34
6.4	ML – Benchmark Analysis	39
7	Discussion	41
7.1	Main Findings	41
7.2	Connection to Existing Literature	42
7.3	Implications for Mental Health Research	42
7.4	Strengths and Limitations	43
7.5	Ethics	43
7.6	Declaration of AI	43
8	Conclusions and Future Work	44
8.1	Conclusions	44
8.2	Future Work	44
	References	46
A	SHAP Clustering	49
B	DCMFNet Model Training	52
B.1	Hyperparameters	52
C	Results	53
C.1	Random Effects	53

List of Abbreviations

PRS	Polygenic Risk Scores
GLMM	Generalized Linear Mixed Model
GAMM	Generalized Additive Mixed Model
DCMFNet	Deep Cross-Modal Fusion Network
GWAS	Genome Wide Association Study
ML	Machine Learning
ACE	Adverse Childhood Experiences
SNP	Single Nucleotide Polymorphism
PCA	Principal Component Analysis
STR	Swedish Twin Registry
CATSS	Child and Adolescent Twin Study in Sweden
LightGBM	Light Gradient Boosting Machine
SHAP	SHapley Additive exPlanations
SUD	Substance Use Disorder
SCZ	Schizophrenia
SES	Socio-Economic Status
ADHD	Attention-Deficit/Hyperactivity Disorder
ASD	Autism Spectrum Disorder
CCA	Complete Case Analysis
MAR	Missing At Random
MNAR	Missing Not At Random
MCAR	Missing Completely At Random
IGF	Iterative Gated Fusion
SE	Squeeze-and-Excitation
ReLU	Rectified Linear Unit
ASGate	Adaptive Selection Gate
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error

List of Figures

3.1	Flow chart of genotype data processing for cell-type PRS generation.	8
3.2	Pearson correlation matrix of cell-type PRS	9
3.3	Feature selection workflow	11
3.4	SHAP-based feature importance for positive and negative outcomes	12
3.5	Distribution of normalised outcome variables	13
4.1	Overview of the adapted DCMFNet architecture for tabular genotype and phenotype data.	16
5.1	Top 10 predictors for positive and negative psychotic symptoms based on GLMM regression coefficients (95% confidence intervals) at seed 42.	19
5.2	Top 10 predictors for positive and negative psychotic symptoms based on GAMM regression coefficients (95% confidence intervals) at seed 42.	21
5.3	Training dynamics for the positive symptom model under default hyperparameters (seed 42).	22
5.4	Training dynamics for the negative symptom model under default hyperparameters (seed 42).	23
5.5	Training dynamics for the positive symptom model after hyperparameter tuning (seed 42).	24
5.6	Training dynamics for the negative symptom model after hyperparameter tuning (seed 42).	24
5.7	Calibration plots for GLMM models (positive and negative outcomes seed=42)	26
5.8	Calibration plots for GAMM models (positive and negative outcomes seed=42)	26
5.9	Calibration plots for DCMFNet models (positive and negative outcomes seed=42)	26
6.1	Main effects in GLMM models for positive and negative outcomes at seed 42 using regression coefficients.	29
6.2	Interaction effects between substance use and covariates in GLMM models, stratified by substance type at seed 42 using regression coefficients.	30
6.3	Main effects in GAMM models for positive and negative outcomes at seed 42 using regression coefficients.	31

6.4	Interaction effects between substance use and covariates in GAMM models at seed 42 using regression coefficients.	32
6.5	Optimal depth search for each modal block in the positive DCMFNet model at seed 42.	33
6.6	Optimal depth search for each modal block in the negative DCMFNet model at seed 42.	33
6.7	Feature importance for independent features in the positive DCMFNet model at seed 42.	35
6.8	Feature importance for independent features in the negative DCMFNet model at seed 42.	35
6.9	Interaction feature importance comparison between high- and low-risk groups (positive model) at seed 42.	37
6.10	Independent feature importance comparison between high- and low-risk groups (positive model) at seed 42.	37
6.11	Interaction feature importance comparison between high- and low-risk groups (negative model) at seed 42.	38
6.12	Independent feature importance comparison between high- and low-risk groups (negative model) at seed 42.	38
6.13	Model performance comparison by algorithm family for positive outcomes. .	39
6.14	Model performance comparison by algorithm family for negative outcomes. .	39
A.1	SHAP Clustering of Positive LightGBM model using Ward linkage	49
A.2	SHAP Clustering of Positive LightGBM model using Complete linkage . . .	50
A.3	SHAP Clustering of Negative LightGBM model using Ward linkage	50
A.4	SHAP Clustering of Negative LightGBM model using Complete linkage . .	51
C.1	Random effect of Twin in GLMM Models at seed 42	53
C.2	Random effect of Twin in GAMM Models at seed 42	53
C.3	Random effect of cell-type PRS smooth terms in Positive GAMM Model . .	55
C.4	Random effect of cell-type PRS smooth terms in Negative GAMM Model .	56

Chapter 1

Introduction

1.1 Background and Motivation

Schizophrenia is a severe psychiatric disorder affecting approximately 1 in 300 individuals worldwide. It is characterized by a combination of positive, negative, and cognitive symptoms. Positive symptoms include hallucinations, delusions, and disorganized thought processes, while negative symptoms reflect a reduction in normal functioning, such as diminished emotional expression and social withdrawal. Cognitive symptoms involve impairments in attention, memory, and executive function. The onset of schizophrenia typically occurs in late adolescence or early adulthood [1]. It is estimated to be 60–80% heritable, making it one of the most strongly genetic mental health disorders [2].

These symptoms arise from disruptions in the underlying neuro-biological systems, particularly involving dopamine signaling across brain regions such as the prefrontal cortex, basal ganglia, and hippocampus [3]. More recent work has shown that these genetic signals are not uniformly distributed across the brain, but are enriched in specific cell types defined by their gene expression profiles [4]. This highlights the importance of considering biological specificity when modeling genetic risk.

Despite this substantial genetic component, genetic predisposition alone does not fully explain the development of the disorder. Twin and family studies demonstrate that many individuals with elevated genetic risk never develop schizophrenia, indicating that environmental and behavioral exposures play a critical role in disease manifestation [2].

Among these exposures, substance use—particularly cannabis—has emerged as one of the most consistently associated risk factors for psychosis. Longitudinal evidence suggests that adolescent cannabis use is linked to an increased risk of psychotic symptoms, especially among individuals with elevated genetic susceptibility [5]. Rather than acting independently, substance use appears to interact with underlying biological vulnerability, reflecting a broader gene–environment interaction framework.

Substance use is therefore treated as the primary exposure in this study. Unlike genetic risk, it is modifiable and varies in timing, frequency, and intensity, particularly during adolescence—a critical period of neuro-development. Since schizophrenia commonly emerges

during late adolescence or early adulthood, exposures during this period may have disproportionate effects on disease risk [5].

Moreover, substance use lies at the intersection of multiple domains. It is shaped by socio-economic conditions, adverse childhood experiences, and behavioral traits, while also influencing neuro-biological processes associated with psychotic symptoms. This makes it a natural focal point for studying how genetic, environmental, and behavioral factors jointly contribute to schizophrenia risk.

In addition to substance use, several developmental and environmental factors have been associated with schizophrenia. Adverse childhood experiences (ACE), including emotional, physical, and sexual abuse, have been linked to increased risk of later psychopathology [6]. Longitudinal studies further indicate that peer bullying and cannabis use jointly contribute to increasing psychotic experiences over time [7].

Neuro-developmental conditions such as autism spectrum disorder and attention-deficit/hyperactivity disorder are also associated with elevated risk. Individuals with autism are three to six times more likely to develop schizophrenia [8], while individuals with ADHD have a relative risk of 4.3 [9]. Socio-economic status appears to act both as a cause and consequence of schizophrenia [10]. Additionally, sex differences have been observed, with men exhibiting more severe negative symptoms and women showing higher levels of depressive symptoms and functioning [11].

1.2 Research Gap

Although substantial research has examined genetic, environmental, and behavioral risk factors for schizophrenia, fewer studies have investigated how these factors jointly interact during adolescence within a unified analytical framework.

In particular, the combined influence of adolescent substance use, adverse childhood experiences, socio-economic status, and genetic predisposition on psychotic symptoms remains incompletely understood. These relationships are inherently complex and are likely to involve non-linear and higher-order interactions across multiple domains.

This complexity is further amplified in population-based cohort data, where variables are high-dimensional, partially correlated, and measured across different developmental stages. In such settings, the number of plausible interactions grows rapidly, and the true structure of these relationships is typically unknown a priori.

As a result, traditional modeling approaches face important limitations. Regression-based models require assumptions about the functional form of relationships and the structure of interactions, which may be difficult to justify in the presence of complex, multi-domain data. This creates a tension between interpretability and flexibility: simpler models may fail to capture relevant patterns, while more flexible models are harder to constrain and interpret.

Together, these challenges highlight the need for modeling approaches that can both accommodate the structure of the data and capture complex interactions without relying

on strong prior assumptions.

1.3 Research Aim and Questions

Scientific aim — How does adolescent substance use at age 15 interact with genetic risk, proxied by polygenic risk scores (PRS), to influence psychotic symptoms in early adulthood? We consider potential confounders including adverse childhood experiences at ages 15 and 18, baseline mood symptoms at age 15, socio-economic status, sex, early ADHD and autism symptoms at age 9, and substance use at age 18.

Methodological aim — Can flexible modeling approaches capture the complex and non-linear interactions between these factors more effectively than traditional regression-based methods?

The following chapter introduces the theoretical and methodological frameworks used to address these questions, including statistical, machine learning, and deep learning approaches.

Chapter 2

Theory

2.1 Genetic Risk and Cell-type Polygenic Risk Scores

Polygenic risk scores (PRS) provide a way to summarize an individual’s genetic susceptibility to a disorder by aggregating the effects of many genetic variants across the genome. In their standard form, PRS are constructed as weighted sums of risk alleles derived from GWAS summary statistics. Although effective as predictors, conventional PRS are often limited in their interpretability, as they combine signals across the entire genome without distinguishing between underlying biological mechanisms.

To address this limitation, cell-type-specific PRS have been developed to incorporate biological context into genetic risk modeling. This approach leverages gene expression data to identify sets of genes that are specifically active in particular types of brain cells. By restricting PRS calculations to variants associated with these gene sets, it becomes possible to construct risk scores that reflect the specific contributions of cell-types to disease [4].

The gene sets used in this study are derived from the human brain cell atlas [12], which characterizes neuronal populations based on transcriptomic profiles. They represent the top 10% most specifically expressed genes for each cell type enriched for schizophrenia-associated genetic variants. These include excitatory and inhibitory neurons in cortical regions, hippocampal sub-fields involved in memory, amygdala neurons associated with emotional processing, and subcortical populations involved in reward and motor function.

Incorporating cell-type PRS allows the analysis to move beyond a purely predictive framework towards a more mechanistic perspective. It enables the investigation of whether specific biological systems are more strongly associated with psychotic symptoms, and how these genetic signals interact with environmental exposures such as substance use.

2.2 Statistical Modeling Approaches

A natural starting point for analyzing such data is the class of regression models, which aim to quantify relationships between predictors and an outcome variable.

Generalized Linear Models (GLM) extend linear regression by allowing for different

types of outcome distributions. However, standard GLMs assume that all observations are independent, which is not appropriate in settings such as twin studies, where individuals are naturally clustered.

Generalized Linear Mixed Models (GLMM) address this limitation by incorporating random effects, which account for correlations within groups, such as twin pairs. In essence, GLMM allows each group to have its own baseline level while estimating overall effects across the population. These models are well-suited for hierarchical data but typically rely on assumptions about linear relationships between predictors and outcomes.

To introduce greater flexibility, Generalized Additive Mixed Models (GAMM) extend GLMM by allowing certain effects to be modeled as smooth, non-linear functions rather than fixed linear terms. This enables the model to capture more complex patterns in the data while still retaining a degree of interpretability. However, in both GLMM and GAMM, interactions between variables must be specified explicitly, and the form of these interactions is determined by the researcher.

2.3 Machine Learning and Deep Learning Approaches

As the number of predictors and potential interactions increases, explicitly specifying all relevant relationships becomes increasingly challenging. Machine learning approaches provide an alternative by learning patterns directly from the data, rather than relying on pre-defined functional forms.

In this context, machine learning refers to a class of algorithms that optimize predictive performance by identifying patterns in the data through training. These models can capture non-linear relationships and interactions implicitly, making them particularly useful in high-dimensional settings. However, they often operate as “black-box” models, meaning that the learned relationships are not easily interpretable.

Deep learning represents a subset of machine learning based on neural networks with multiple layers. These models learn hierarchical representations of data, where simple patterns are combined to form increasingly complex features. This layered structure enables deep learning models to capture high-order interactions and complex dependencies between variables.

Machine learning and deep learning approaches are increasingly explored in mental health research for integrating high-dimensional and multi-domain data. For instance, prior work using CATSS data applied several machine learning models to predict adolescent mental health outcomes, demonstrating their potential while showing comparable performance to traditional regression approaches [13].

This suggests that while these methods can capture complex relationships, their practical advantage depends on the underlying structure of the data.

Among deep learning architectures, models designed for cross fusion of multimodal data are particularly relevant in this study. The Deep Cross-Modal Fusion Network (DCMFNet) was originally proposed to integrate heterogeneous data sources, such as images and text,

by iteratively combining information across modalities [14]. The core idea is to allow different groups of features to interact in a structured manner, rather than treating all variables as a single homogeneous set.

In this study, all the genetic, environmental, and behavioral variables are treated as distinct modalities within the DCMFNet framework. This allows the model to learn how these domains interact with one another, capturing complex and potentially non-linear relationships that are difficult to specify explicitly in traditional models.

To evaluate the effectiveness of this approach, it is compared with traditional statistical models, including Generalized Linear Mixed Models (GLMM) and Generalized Additive Mixed Models (GAMM).

To contextualize the model performances, we include a range of standard machine learning models implemented using the `lazypredict` package [15], which enables rapid evaluation of multiple regression algorithms under a consistent framework. This includes models from different families such as linear, regularized, tree-based, ensemble, kernel, and nearest-neighbor methods.

These models were trained without hyperparameter tuning to provide an out-of-the-box benchmark of predictive performance. These models serve as empirical benchmarks rather than objects of interpretation, as their interactions are learned implicitly and cannot be explicitly specified, making direct comparison with structured models such as GLMM, GAMM, and DCMFNet less meaningful.

The primary focus of this study is on comparing interpretable statistical models with flexible deep learning approaches in modeling complex interactions. To enable such a comparison, it is necessary to first establish a coherent and well-structured dataset that reflects the interplay of genetic, environmental, and behavioral factors over time. The following chapter therefore outlines the study cohort, the derivation of genetic risk measures, and the pre-processing steps undertaken to prepare the data for analysis.

Chapter 3

Data

3.1 Study Cohort

This study uses data from the Swedish Twin Registry (STR), specifically the Child and Adolescent Twin Study in Sweden (CATSS), which provides cross-sectional questionnaire-based data at pre-defined ages 9, 15, and 18. This design enables the analysis of developmental relationships between early-life exposures and later mental health outcomes.

The primary exposure is substance use at age 15, and the outcomes are psychotic and affective symptoms at age 18. Positive symptoms include psychotic and manic features, while negative symptoms are proxied by depressive symptoms.

3.2 Genotype Data and Cell-type PRS Generation

Genotype data for CATSS individuals were obtained from the STR, and cell-type PRS were computed using the PRS-CS framework [16]. The selection of individuals for genotype analysis is summarized in Figure 3.1. This process ensures that only individuals with complete and reliable genotype information are included in downstream analyses.

3.2.1 Cell-type PRS Pipeline

The PRS pipeline, adapted from Wang et al. [17], involves quality control, SNP annotation, genotype cleaning, and PRS computation across chromosomes. Specifically, SNP effect sizes are estimated using a Bayesian framework (PRS-CS) ¹ and aggregated to generate individual-level risk scores for each cell type. This systematic pipeline ensures that the resulting PRS are robust, biologically informed, and comparable across individuals.

Correlation analysis of the resulting PRS revealed clustering patterns across neuronal groups. This is visualized in Figure 3.2, where excitatory, inhibitory, and spiny neuron PRS show within-group correlation structures, suggesting shared genetic architecture. Overall, moderate positive correlations are observed across most cell-type PRS, reflecting a shared polygenic architecture underlying schizophrenia risk. The relatively stronger correlations

¹<https://github.com/getian107/PRScs/tree/master>

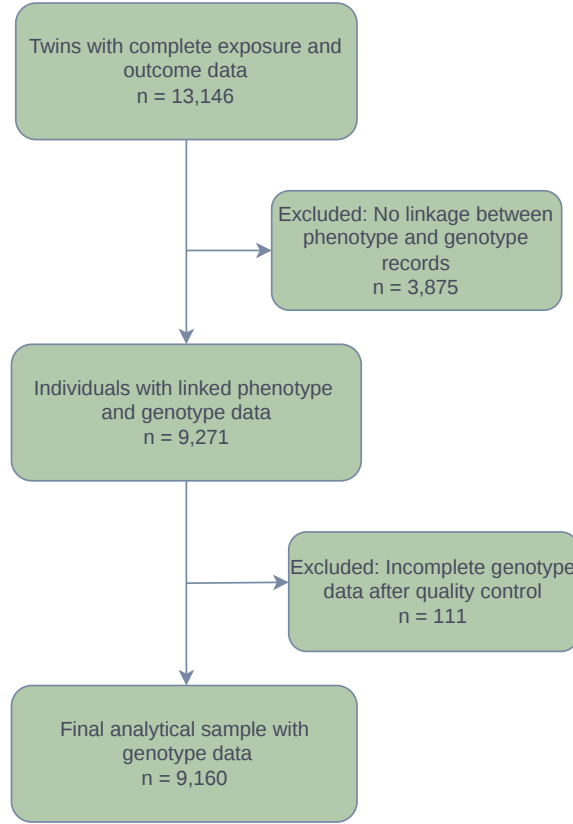


Figure 3.1: Flow chart of genotype data processing for cell-type PRS generation.

within biologically related neuronal groups suggest that these scores capture overlapping functional pathways. Although, no pairwise correlation exceeds 0.7, indicating that strong multicollinearity is not present among the predictors. This supports the inclusion of all cell-type PRS in subsequent analyses, as each score is likely to contribute complementary information without compromising model stability.

Genetic confounding: To account for population structure and batch effects, principal component analysis (PCA) was applied. PRS were adjusted using batch-specific principal components, and individuals with non-European ancestry were identified as potential outliers.

3.3 Phenotype Data

The phenotype data includes behavioral, environmental, and clinical variables derived from questionnaires across multiple time points. The dataset contains over 200 variables, many of which are correlated due to overlapping constructs and measurement scales.

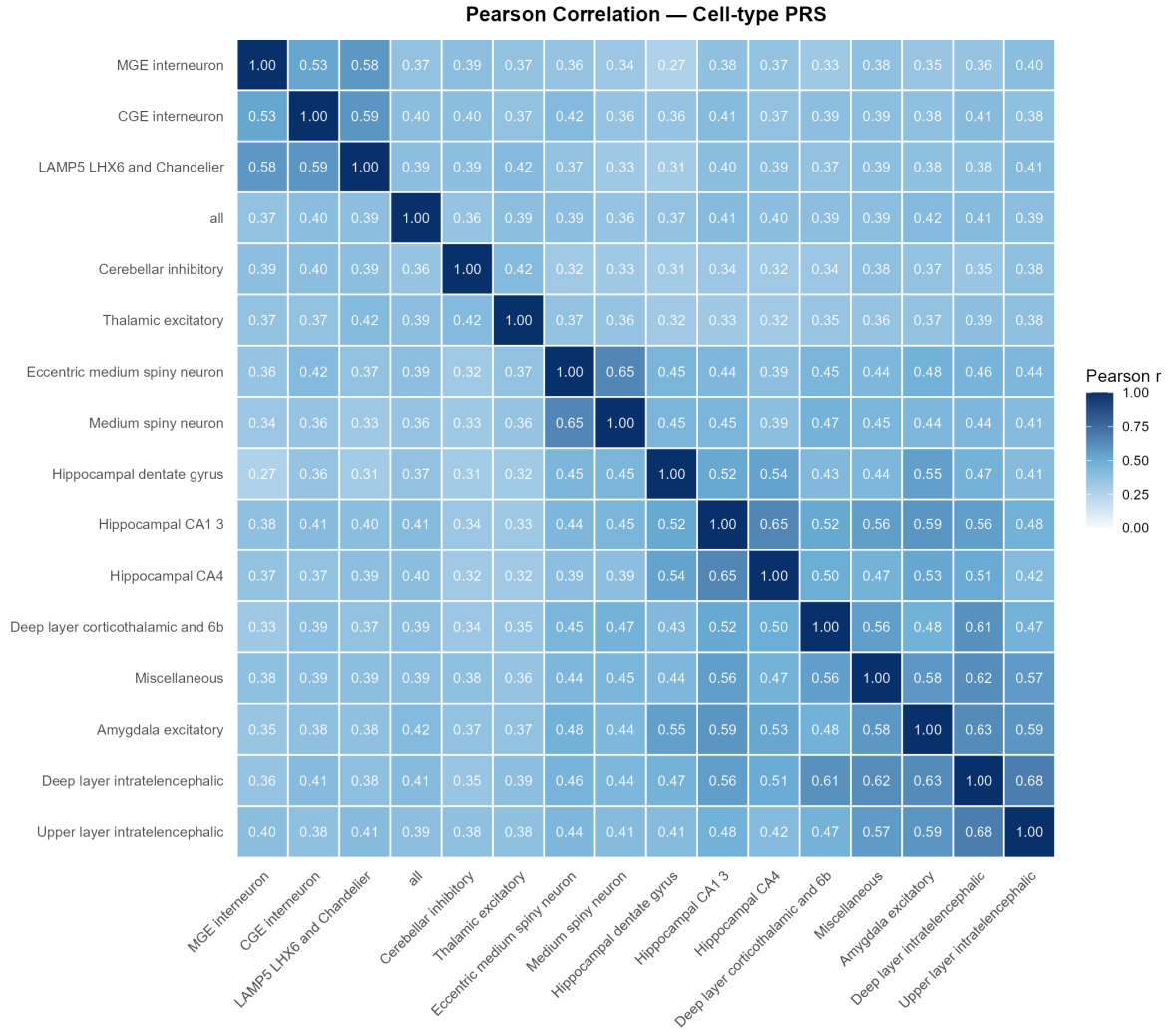


Figure 3.2: Pearson correlation matrix of cell-type PRS

The variables included in this study span multiple domains:

- **Exposure:** Substance use at age 15, including cannabis, other drugs, smoking, alcohol, and snuff.
- **Environmental factors:** Adverse Childhood Experiences (ACE) at ages 9, 15, and 18, including abuse, bullying, neglect, and exposure to violence, along with early ADHD and autism-related symptoms at age 9.
- **Demographic factors:** Socio-economic status (SES), parental education, country of birth, and sex.
- **Baseline symptoms:** Psychotic, manic, and depressive symptoms at age 15.
- **Outcomes:** Psychotic, manic, and depressive symptoms at age 18.

These variables are derived from structured clinical questionnaires and are inherently high-dimensional, with each construct represented by multiple symptom-level indicators.

Substance use and schizophrenia-related symptoms are both measured at age 15. As the temporal ordering between these variables cannot be established, baseline schizophrenia symptoms (SCZ15) are included as an adjustment variable, acknowledging their potential role as either confounders or mediators.

Initial descriptive analyses were conducted on the full set of variables. However, due to high dimensionality and strong correlations, the results were difficult to present meaningfully. Feature selection was therefore applied to reduce complexity and improve interpretability.

3.4 Feature Selection

Feature selection was conducted as a structured pipeline combining statistical exploration and model-based approaches. First, pairwise Pearson correlations were examined to identify redundancy and inform grouping of related variables. Second, a LightGBM model was used to evaluate feature importance. LightGBM is a gradient boosting framework that combines multiple decision trees to model complex relationships efficiently [18]. Its tree-based structure naturally captures non-linear effects and interactions, making it well-suited for exploratory feature screening in high-dimensional data.

Feature importance was interpreted using SHAP (TreeSHAP) values, which quantify the contribution of each feature to model predictions. SHAP clustering was used to identify groups of variables with similar influence patterns (see Appendix A).

The overall feature selection workflow is summarised in Figure 3.3, which outlines the sequential filtering and grouping steps.

The following criteria were applied for feature selection:

- Removal of variables with very high missingness ($> 90\%$)
- Grouping or removal of highly correlated variables (> 0.9)
- Use of SHAP-based clustering for moderately correlated features
- Removal of variables with 50–90% missingness when not informative

Based on these criteria, highly correlated substance use variables were aggregated, parent-reported outcomes were removed due to higher missingness, and ACE-related variables were grouped to reduce redundancy.

For instance, the `OtherDrugs15` variable aggregates multiple highly correlated substances, including amphetamines, cocaine, opioids (e.g., heroin, morphine), psychedelics (e.g., LSD, mushrooms), inhalants, and performance-enhancing drugs. Outcome variables were constructed as normalized aggregated symptom scores between 0 and 1, preserving variability while avoiding arbitrary thresholds.

Feature importance patterns are visualized in Figure 3.4 using SHAP beeswarm plots, where each point represents the contribution of a feature to an individual prediction.

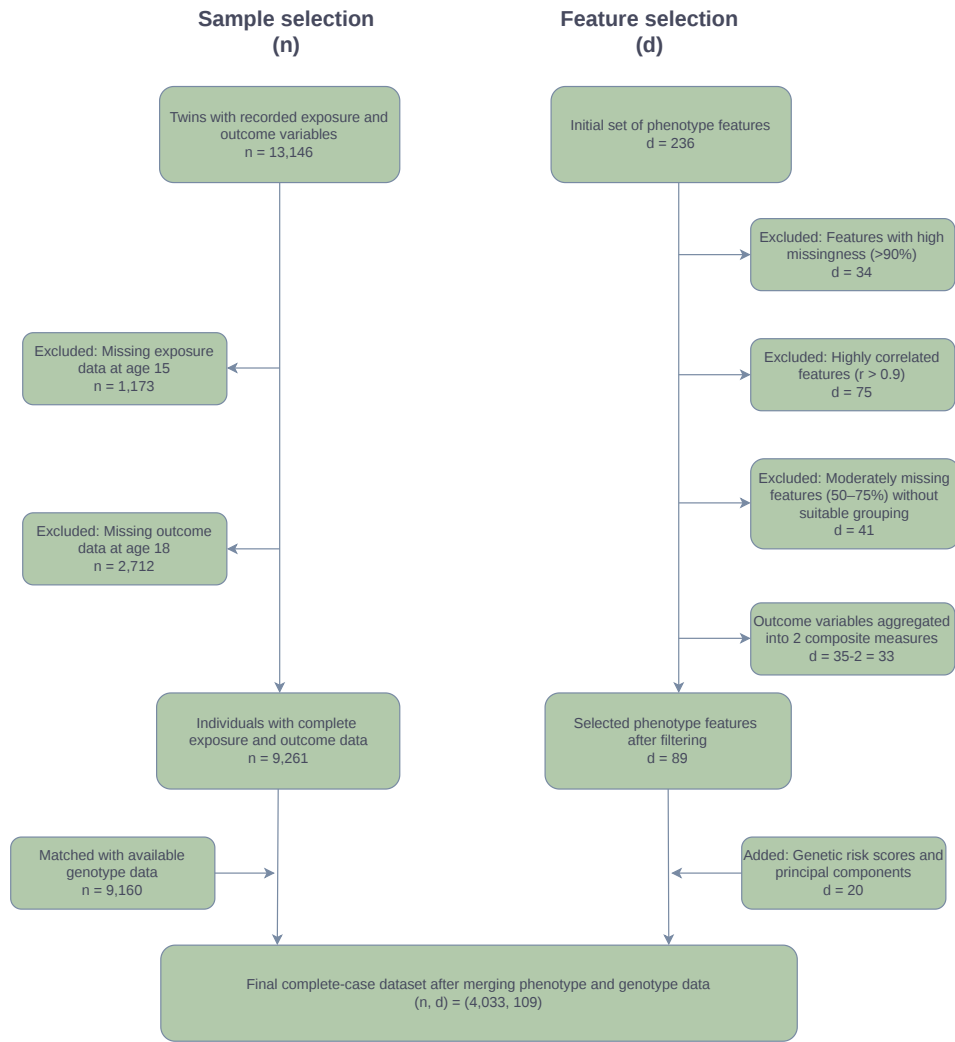


Figure 3.3: Feature selection workflow

The horizontal position indicates whether the feature increases (right) or decreases (left) the predicted outcome, while color reflects the feature value (red = high, blue = low). The spread of points along the x-axis reflects the variability and overall influence of each feature.

Features such as substance use (e.g., cigarettes and snuff at age 18), emotional abuse, and affective symptoms (e.g., low mood, restlessness) show consistent positive contributions to predicted risk. Additionally, baseline psychological features at age 15 (e.g., racing thoughts, worry, confidence) contribute meaningfully, highlighting the role of early mental health signals in later outcomes.

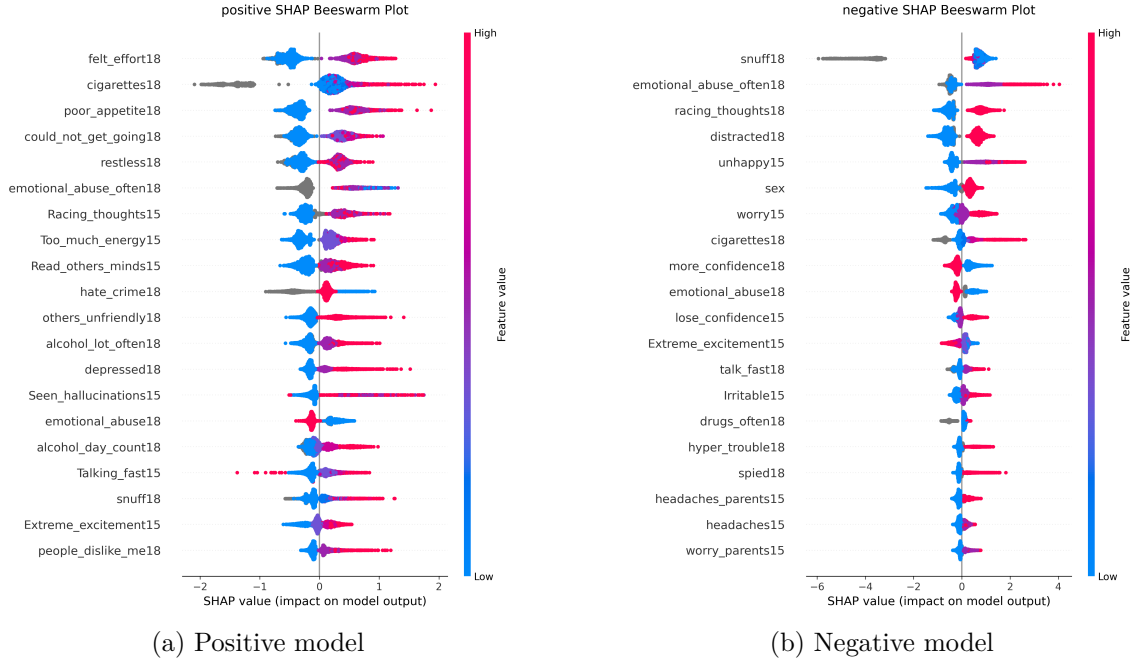


Figure 3.4: SHAP-based feature importance for positive and negative outcomes

3.5 Descriptive Statistics

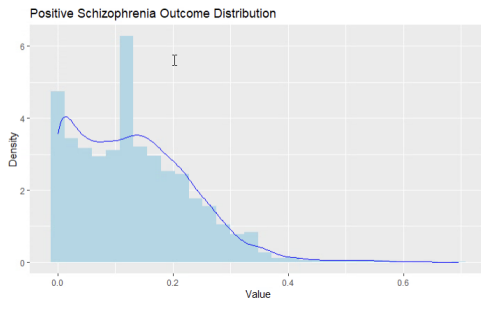
Table 3.1 summarizes the key modalities included in the analysis, highlighting a mix of ordinal, binary, and continuous variables spanning behavioral, environmental, and genetic domains.

Overall, most variables exhibit low to moderate missingness, although earlier-life measures such as SCZ15 and ADHD9 show higher proportions of missing data, reflecting challenges in data collection over the years.

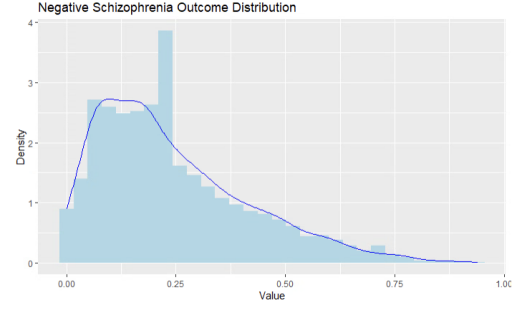
The outcome variables (SCZ18 positive and negative symptoms) are continuous and skewed towards lower values, consistent with the low prevalence of high-risk cases observed in the cohort, as also seen in Figure 3.5.

Modality	N Vars	Type	Mean $\pm$ SD	Median	Range	Missing
SUD15 (Substance Use, age 15)	6	Ordinal	0.706 $\pm$ 1.388	0.000	0.000–5.000	0.0%
PRS (Polygenic Risk Scores)	16	Continuous	-0.085 $\pm$ 0.276	-0.044	-2.648–0.668	28.3%
SCZ15 (Psychotic Experiences, age 15)	23	Ordinal	0.448 $\pm$ 0.733	0.000	0.000–3.000	27.4%
ADHD9 (ADHD Symptoms, age 9)	19	Ordinal	0.160 $\pm$ 0.443	0.000	0.000–2.000	14.9%
ASD9 (ASD Traits, age 9)	17	Ordinal	0.071 $\pm$ 0.303	0.000	0.000–2.000	15.9%
ACE15 (Adverse Experiences, age 15)	7	Ordinal	1.219 $\pm$ 0.693	1.000	1.000–6.000	4.6%
ACE18 (Adverse Experiences, age 18)	4	Binary	88.7%	NA	0, 1	11.1%
SUD18 (Substance Use, age 18)	4	Ordinal	0.925 $\pm$ 1.479	0.000	0.000–7.000	0.3%
SES (Socioeconomic Status)	4	Ordinal	2.550 $\pm$ 1.097	2.000	0.000–5.000	18.0%
SEX	1	Categorical	Male: 3221 (34.8%), Female: 4683 (50.6%)	NA	1, 2	14.7%
SCZ18 Positive (Outcome)	1	Continuous	0.139 $\pm$ 0.106	0.130	0.000–0.870	0.0%
SCZ18 Negative (Outcome)	1	Continuous	0.256 $\pm$ 0.181	0.212	0.000–1.000	0.0%

Table 3.1: Descriptive statistics of input modalities and outcome variables.



(a) Positive symptoms



(b) Negative symptoms

Figure 3.5: Distribution of normalised outcome variables

3.6 Handling Missing Data

For the covariates in this data, the missingness mechanism could not be clearly classified as MCAR, MAR, or MNAR. Variables such as ACE9 and baseline symptoms (SCZ15) are temporally prior to most other covariates, leaving insufficient information for reliable imputation without introducing strong assumptions. Similarly, parental-reported variables exhibited high missingness and could not be reconstructed from available data. To preserve the temporal structure and avoid potential bias, variables with substantial missingness were not imputed and were excluded where appropriate.

Sparse responses such as `Don't know`/`Don't want to answer` were treated as missing values to maintain consistency.

Chapter 4

Methodology

4.1 Modeling Strategy

Two separate models were constructed for the two outcomes of interest: positive and negative psychotic symptoms at age 18. Each modeling approach was applied consistently across both outcomes to ensure comparability.

The dataset was split into training and testing sets using a 75–25 split, while preserving the distribution of the outcome variables. To avoid data leakage and account for the twin structure, both individuals from a twin pair were always assigned to the same split.

4.2 Statistical Models

4.2.1 Generalized Linear Mixed Model (GLMM)

We first implemented a GLMM with psychotic symptoms as the outcome variable. All environmental, genetic, and behavioral variables were included as covariates. The outcome was modeled using a Gaussian distribution with an identity link, consistent with the continuous nature of the constructed symptom scores.

To explicitly study interactions with the primary exposure, interaction terms between SUD15 and all other covariates were included. Genetic confounding was accounted for by including *batch* $\times$ *PC* terms. The twin structure was incorporated through a random intercept for each twin pair.

The model can be expressed as:

$$\text{SCZ18} \sim \text{covariate_terms} + \text{interaction_terms} + \\ \text{genetic_confounding_terms} + (1 \mid \text{cmpair})$$

This formulation provides a structured and interpretable baseline, allowing us to estimate both main effects and pre-specified interactions while accounting for within-pair dependence in the twin data.

4.2.2 Generalized Additive Mixed Model (GAMM)

To relax the assumption of linearity in the GLMM, we implemented a GAMM. The model specification is similar to the GLMM, but allows for non-linear relationships through smooth functions, applied here to each cell-type PRS variable to avoid assuming a strictly linear relationship between genetic risk and psychotic outcomes.

The model is specified as:

$$\text{SCZ18} \sim \text{covariate_terms} + s(\text{PRS}_k) + \text{interaction_terms} + \\ \text{genetic_confounding_terms} + s(\text{cmpair}, bs = 're')$$

Taken together, GLMM and GAMM provide complementary statistical baselines: the former offers a strictly parametric framework, while the latter introduces limited flexibility through smooth functions and extends to capture moderate non-linear effects.

While, the GAMM relaxes linearity for individual PRS terms, it cannot capture higher-order interactions between covariates or learn latent representations of combined exposures — motivating the use of a deep cross-modal fusion approach.

4.3 Deep Learning Model

4.3.1 DCMFNet Architecture

The Deep Cross-Modal Fusion Network (DCMFNet) is designed to explicitly model interactions between a primary exposure and multiple covariates through iterative cross-feature fusion. In this study, the architecture consists of 9 Iterative Gated Fusion (IGF) blocks, where each block models the interaction between the primary exposure (*SUD15*) and a single covariate. The overall architecture is illustrated in Figure 4.1.

Within each IGF block, the guided feature (*SUD15*) and a guidance feature (covariate) are fused iteratively across multiple layers (set to 5 by default, following [14]). This process progressively refines the interaction representation through repeated transformation and filtering.

The fusion update at each layer is defined as:

$$F_{curr} = \tau(\tau(W_1 x) \odot \tau(W_2 G_{prev}))$$

where τ denotes the tanh activation, $\odot$ represents element-wise multiplication, and $W_1 \in \mathbb{R}^{N_x \times N_m}$ and $W_2 \in \mathbb{R}^{N_m \times N_m}$ are learnable weight matrices. Here, x corresponds to the guided feature (*SUD15*), and $G_{prev} \in \mathbb{R}^{b \times N_m}$ is the output of the gating mechanism from the previous layer, with b denoting the batch size.

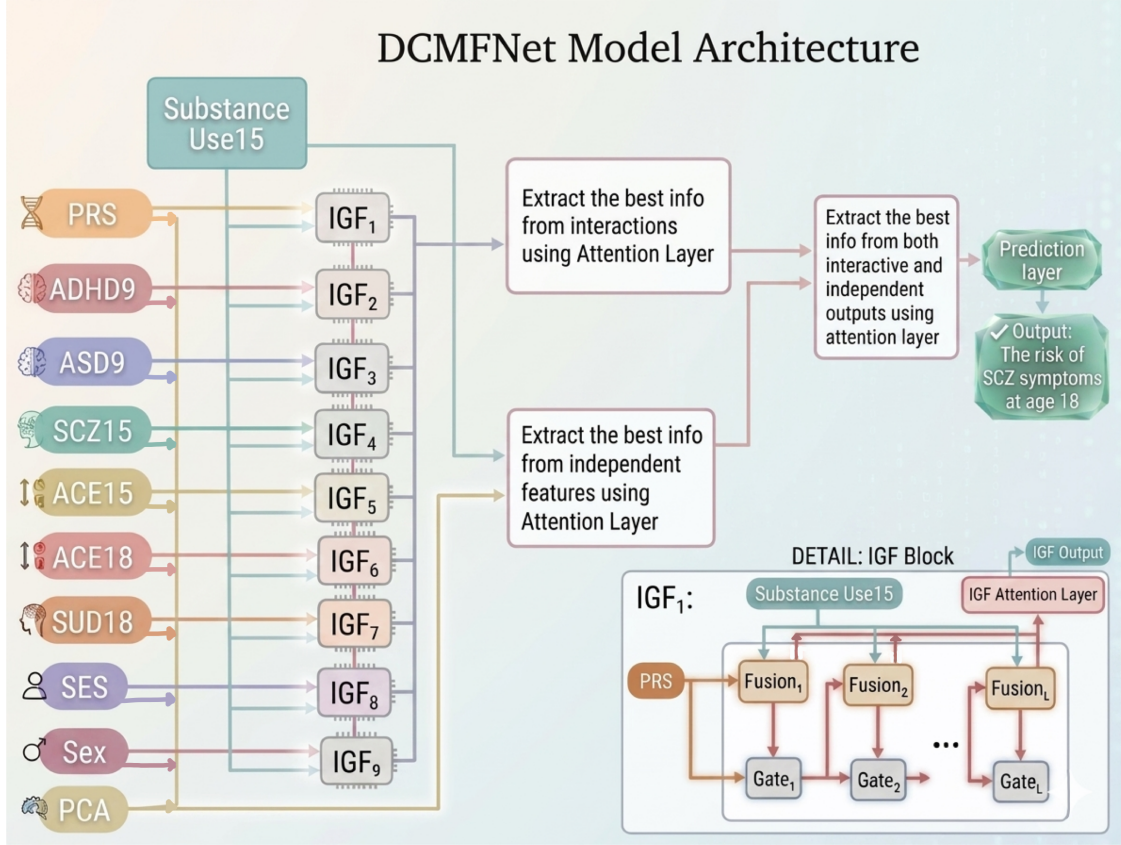


Figure 4.1: Overview of the adapted DCMFNet architecture for tabular genotype and phenotype data.

To mitigate noise introduced during iterative fusion, an adaptive selection gate (AS-Gate) is applied:

$$z_t = \sigma(W_z(F_t + G_{t-1}) + b_z)$$

$$f_t = F_t \odot z_t$$

$$G_t = \tau(W_g([F_t; f_t]) + b_g)$$

where $W_z \in \mathbb{R}^{N_m \times N_m}$ and $W_g \in \mathbb{R}^{2N_m \times N_m}$ are learnable parameters, b_z and b_g are bias terms, and σ denotes the sigmoid activation.

This gating mechanism selectively amplifies informative signals while suppressing noise, which is particularly important in high-dimensional settings where spurious interactions may otherwise dominate the learned representations.

4.3.2 Model Adaptation

Attention Mechanism

Tabular data lacks inherent spatial or sequential structure, making convolution-based operations unsuitable. We therefore employ a channel-wise attention mechanism to capture dependencies between features.

We adopt the Squeeze-and-Excitation (SE) mechanism [19], which recalibrates feature representations by modeling inter-feature relationships. For an intermediate representation $X \in \mathbb{R}^{b \times C}$, where b is the batch size and C the number of features, the SE block operates as follows:

Squeeze: A global summary is computed for each feature:

$$z_c = \frac{1}{b} \sum_{i=1}^b X_{i,c}$$

Excitation: Feature dependencies are captured via a gating function:

$$s = \sigma(W_2 \cdot \delta(W_1 \cdot z))$$

where W_1 and W_2 are learnable parameters and δ denotes a non-linear activation (ReLU).

Recalibration: The features are rescaled as:

$$\tilde{X}_{i,c} = s_c \cdot X_{i,c}$$

This mechanism emphasizes informative features while suppressing less relevant ones. Unlike soft attention, which assigns weights at the instance level, SE attention captures global feature importance, leading to improved stability and reduced overfitting in this setting.

Loss Function

The outcome variables exhibit a highly skewed distribution, with a small number of high-risk observations. To address this imbalance in a regression setting, we employ a composite loss function that combines robust error estimation with adaptive reweighting.

Base Loss (Huber Loss): The Huber loss is defined as:

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & \text{if } |y - \hat{y}| \leq \delta \\ \delta \left(|y - \hat{y}| - \frac{1}{2}\delta \right), & \text{otherwise} \end{cases}$$

This loss behaves like Mean Squared Error (MSE) for small residuals and Mean Absolute Error (MAE) for large residuals, providing robustness to outliers.

Inverse-Frequency Weighting: Each observation is assigned a weight based on the inverse frequency of its target bin:

$$w_i = \frac{1}{\text{freq}(y_i)}$$

This ensures that rare high-value outcomes contribute more to the loss.

Focal Regression Component: To further emphasize difficult-to-predict observations, a focal weighting term is applied:

$$L_{focal} = w_i \cdot |y_i - \hat{y}_i|^\gamma \cdot L_\delta(y_i, \hat{y}_i)$$

where $\gamma > 1$ controls the degree of emphasis on larger errors. This formulation prioritizes samples with higher residuals, which are often underrepresented tail cases.

The final loss is computed as the weighted sum across all samples.

4.4 Model Training and Evaluation

All models were trained using the same train–test split to ensure a fair comparison. Model performance was evaluated on the test set using residual R^2 , Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Pearson correlation (r), and Spearman correlation (ρ).

Residual R^2 quantifies the proportion of variance explained by the model, RMSE and MAE measure the magnitude of prediction error (with RMSE penalizing larger errors more strongly), while Pearson r and Spearman ρ assess linear and rank-based agreement between predicted and observed values, respectively. In particular, correlation-based metrics are included to assess whether the models capture the relative ordering of risk, which is important in the presence of skewed outcome distributions.

Together, these metrics provide complementary perspectives on predictive performance, but do not assess causal relationships, model calibration, or the correctness of inferred interaction structures.

The following chapter presents the results and evaluates the relative performance of the proposed modeling approaches.

Chapter 5

Results

5.1 GLMM

We first examine the results from the GLMM for both positive and negative psychotic symptom outcomes.

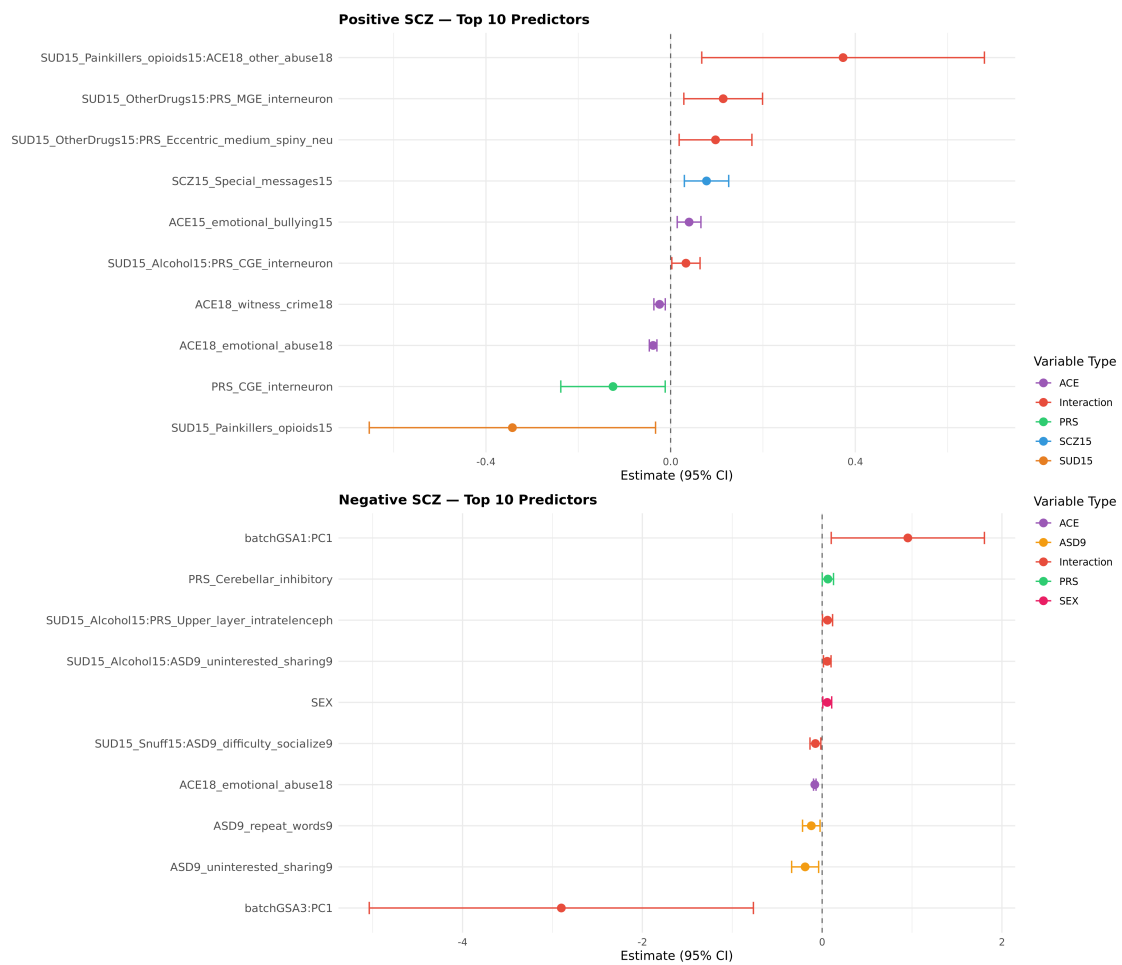


Figure 5.1: Top 10 predictors for positive and negative psychotic symptoms based on GLMM regression coefficients (95% confidence intervals) at seed 42.

Figure 5.1 presents the top predictors ranked by effect size. For positive symptoms, interaction terms between substance use and environmental or genetic factors dominate the top predictors, indicating that risk is primarily driven by combined exposures rather than isolated effects. In particular, interactions involving substance use and adverse experiences or cell-type PRS show the strongest positive associations. Importantly, most confidence intervals for the top predictors do not cross zero, indicating statistically meaningful associations.

In contrast, the negative symptom model exhibits a different structure, with comparatively stronger contributions from genetic components and confounding terms (e.g., batch $\times$ PC), alongside smaller interaction effects. This suggests that negative symptoms may be less driven by acute exposure interactions and more influenced by underlying biological or background variation.

Across both outcomes, several variables show consistent directional effects. Adverse childhood experiences (e.g., emotional abuse at age 18) and baseline symptom measures at age 15 (e.g., energy levels, headaches, rage attacks) are positively associated with later symptom severity, indicating their role as stable risk indicators. The genome-wide PRS (*PRS_all*) also shows consistent effects across both models.

However, some features exhibit outcome-specific behavior. Notably, extreme excitement at age 15 displays opposite directions in the positive and negative models, suggesting that certain early behavioral traits may differentially relate to distinct symptom dimensions.

Model diagnostics further support the validity of the mixed-effects specification. QQ plots of the twin-level random effects (see Appendix C.1) indicate approximate normality, suggesting that within-pair dependence is appropriately captured by the random intercept structure.

5.2 GAMM

We next examine the results from the GAMM for both positive and negative psychotic symptom outcomes.

Figure 5.2 presents the top predictors ranked by effect size. Similar to the GLMM results, interaction terms between substance use and environmental or genetic factors remain among the strongest predictors for positive symptoms. In particular, interactions involving opioid use and adverse experiences, as well as substance use with cell-type PRS, continue to exhibit large positive effects, indicating robust interaction-driven risk patterns.

However, compared to the GLMM, the GAMM allows for non-linear effects, particularly in the PRS. This results in a slight redistribution of effect sizes, with some genetic components appearing more prominently, suggesting that the relationship between genetic risk and psychotic symptoms is not strictly linear.

For negative symptoms, the overall structure remains consistent with the GLMM, with genetic confounding terms (e.g., batch $\times$ PC) and PRS components playing a dominant

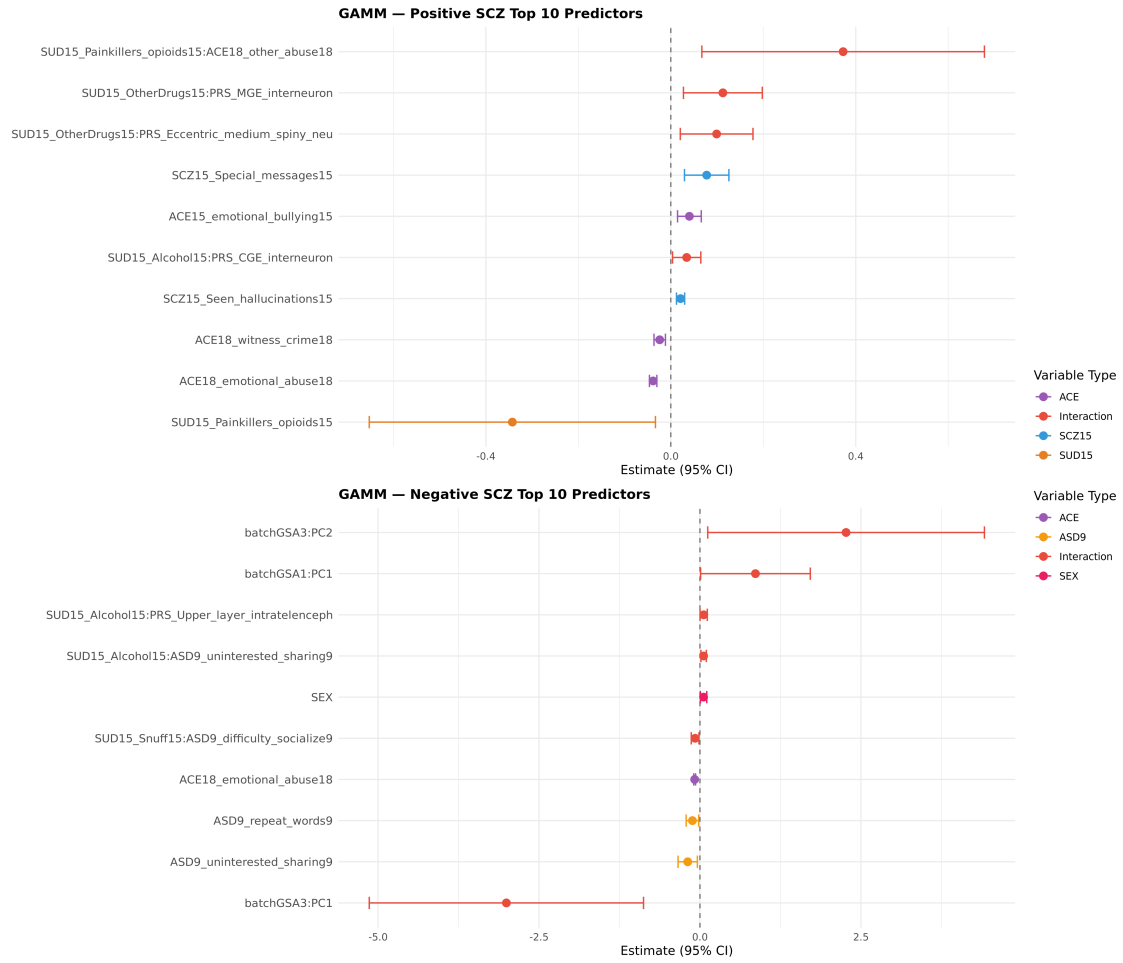


Figure 5.2: Top 10 predictors for positive and negative psychotic symptoms based on GAMM regression coefficients (95% confidence intervals) at seed 42.

role. Interaction terms involving substance use and neuro-developmental traits (e.g., ASD-related variables) are also present, but with relatively smaller magnitudes.

Across both outcomes, several predictors show consistent directional effects. Adverse childhood experiences (e.g., emotional abuse at age 18) and baseline symptom measures at age 15 (e.g., energy levels, headaches, rage attacks) are positively associated with later outcomes, reinforcing their role as stable risk factors. Conversely, extreme excitement at age 15 again shows opposite directions across positive and negative symptom models, indicating differential associations across symptom domains.

Model diagnostics further support the validity of the GAMM specification. QQ plots of the twin-level random effects (see Appendix C.1) indicate approximate normality, suggesting that the mixed-effects structure appropriately captures within-pair dependence. Additionally, non-linear effects of PRS components were generally weak, suggesting that their contribution is better captured through interactions rather than independent smooth terms.

Compared to GLMM, these results suggest that allowing for non-linear effects does not

materially change the dominant patterns identified by the GLMM, although it allows small deviations from linearity. Most smooth terms remain weak and close to linear, indicating limited independent contribution of non-linear PRS effects.

5.3 DCMFNet

5.3.1 Default Model Training

The DCMFNet model was first trained using the default hyperparameters reported in [14]. An internal 80–20 train–validation split was used during training, and the final model was evaluated on the held-out test set.

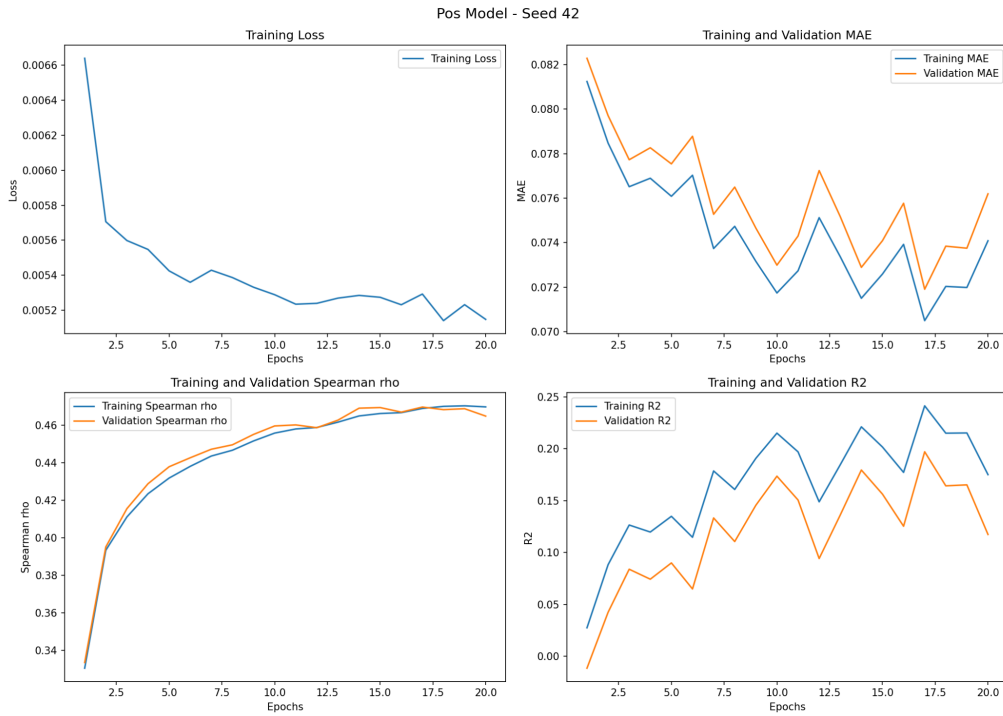


Figure 5.3: Training dynamics for the positive symptom model under default hyperparameters (seed 42).

Across both positive and negative symptom models as seen in Figure 5.3 and 5.4, training loss decreases rapidly in the initial epochs and then stabilizes, indicating effective optimization. Validation MAE follows a similar trend, with a small but consistent gap relative to training MAE, suggesting mild but controlled overfitting.

Spearman correlation increases steadily during training for both models, reaching approximately 0.46–0.50, indicating that the model is able to capture meaningful rank-based structure in the outcomes. Similarly, R^2 improves over epochs but remains modest (approximately 0.15–0.20 on validation), reflecting limited explained variance despite stable learning.

Notably, validation curves exhibit fluctuations across epochs, particularly for R^2 and

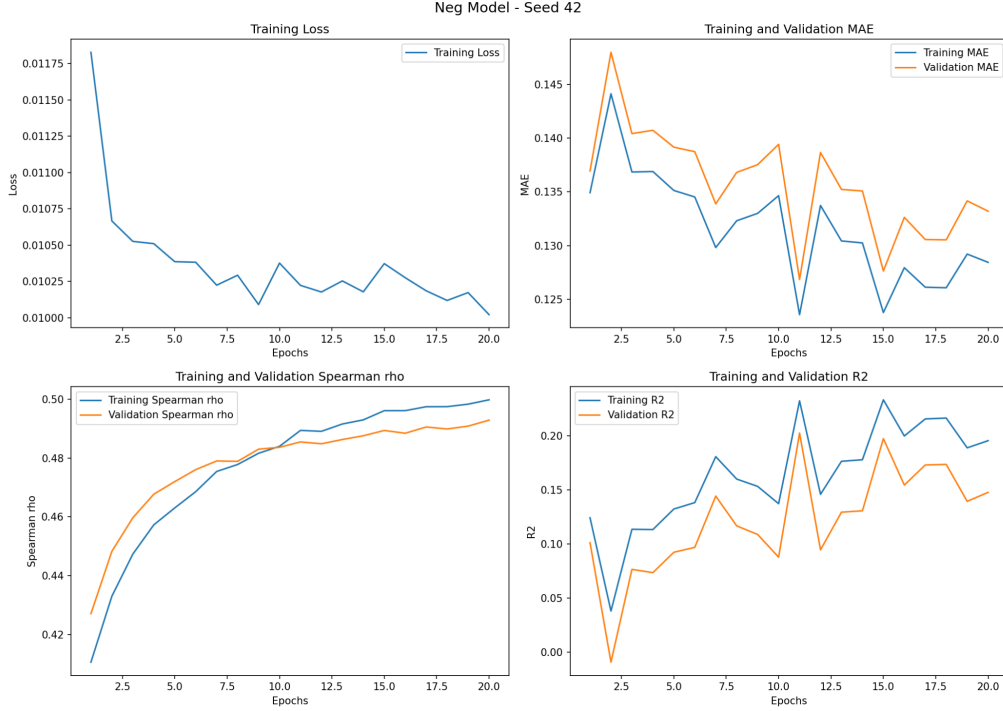


Figure 5.4: Training dynamics for the negative symptom model under default hyperparameters (seed 42).

MAE, indicating sensitivity to the default hyperparameter configuration. This suggests that while the model converges, its generalization performance is not fully optimized, motivating further hyperparameter tuning.

5.3.2 Hyperparameter Tuning

To improve model generalization, hyperparameters were optimized using Optuna, including the number of IGF layers. Optuna performs efficient hyperparameter search using Bayesian optimization, iteratively selecting promising configurations based on past evaluations. A total of 50 trials were conducted, where each trial corresponds to a different hyperparameter configuration evaluated on the validation set. This was necessary to avoid biasing the search toward configurations optimal only for the default architecture (five layers), and instead allow the model complexity itself to be learned from the data.

Regularization strategies were incorporated to improve generalization and stabilize training. A learning rate scheduler was used to gradually decrease the step size during optimization, allowing faster convergence in early epochs and finer updates later. Early stopping was applied based on validation performance to prevent overfitting, halting training once improvements plateaued. Additionally, weight decay was introduced as an ℓ_2 penalty on model parameters to discourage overly complex representations and reduce variance. Together, these strategies aim to balance model flexibility with robustness in a relatively small and high-dimensional dataset.

The default and optimized hyperparameter configurations are provided in Appendix B.1, highlighting differences in model complexity and regularization between the two outcomes. The attention mechanism within each IGF block was excluded from tuning due to overfitting and fixed at its default configuration, while the global attention layers aggregating interaction- and feature-level representations were tuned.

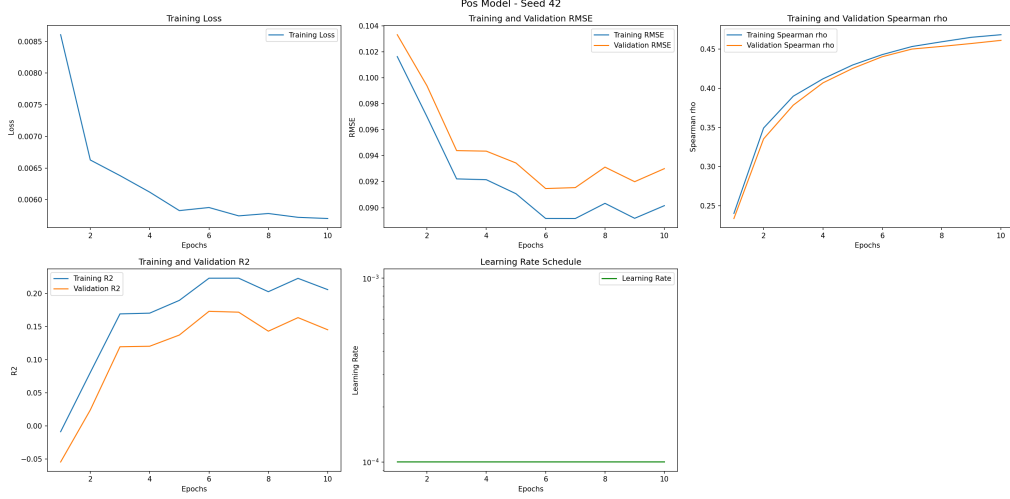


Figure 5.5: Training dynamics for the positive symptom model after hyperparameter tuning (seed 42).

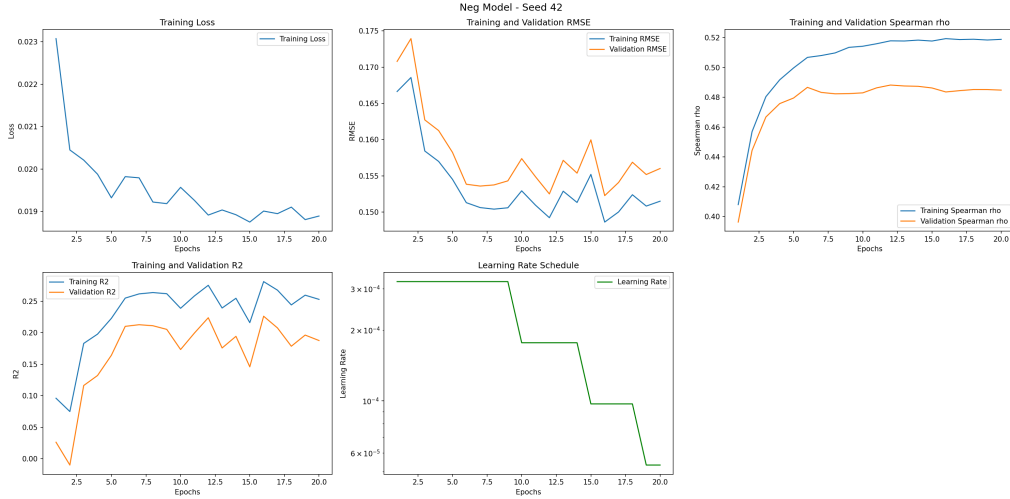


Figure 5.6: Training dynamics for the negative symptom model after hyperparameter tuning (seed 42).

Compared to the default configuration, the tuned models exhibit smoother and more stable training dynamics across all metrics as seen in Figure 5.5 and 5.6. Training and validation RMSE decrease consistently and remain closely aligned, indicating improved generalization and reduced overfitting.

Spearman correlation improves steadily with minimal divergence between training and

validation curves, suggesting that the model captures robust rank-based relationships without overfitting to the training data. Notably, the tuned models achieve comparable or slightly improved correlation levels relative to the default setting, but with greater stability across epochs.

Similarly, validation R^2 shows more consistent improvement, with reduced fluctuations compared to the default model. While the overall magnitude of R^2 remains modest, the reduced variance across epochs indicates a more reliable fit.

The learning rate schedule further supports stable optimization, with stepwise decay enabling initial rapid learning followed by fine-tuning in later epochs.

Overall, hyperparameter tuning leads to more stable convergence and improved generalization, rather than large gains in absolute performance. This suggests that model performance is primarily limited by signal strength in the data rather than optimization inefficiencies. This pattern is consistent with high-dimensional observational data, where improved optimization primarily enhances stability rather than substantially increasing predictive power.

5.4 Model Comparison

Model performance was compared using calibration plots, which illustrate the relationship between predicted and observed values for both positive and negative outcomes (Figures 5.7, 5.8, and 5.9).

Across all models, a general pattern of regression toward the mean is observed, with predictions concentrated in a narrow range relative to the true values. This effect is particularly pronounced in the GLMM and GAMM models, where predictions remain close to the average outcome even at higher observed values.

The GLMM provides a reasonable baseline fit, with moderate alignment between predicted and observed values. The GAMM, despite allowing for non-linear effects, does not yield substantial improvements over the GLMM, suggesting that the added flexibility in modeling smooth effects of PRS does not significantly enhance predictive performance in this setting.

In contrast, the tuned DCMFNet model demonstrates improved performance across both outcomes, as reflected in higher R^2 and correlation metrics. More importantly, the calibration plots indicate that DCMFNet is better able to capture variability in the upper range of the outcome distribution. For the negative symptom model in particular, DCMFNet produces higher predicted values for individuals with elevated observed risk, whereas GLMM and GAMM tend to underestimate these cases.

This difference is critical, as the ability to distinguish higher-risk individuals is essential in this application. While all models perform comparably in the lower range of outcomes, DCMFNet exhibits greater sensitivity to tail behavior, suggesting that it is more effective in capturing complex, non-linear interactions that drive elevated symptom severity.

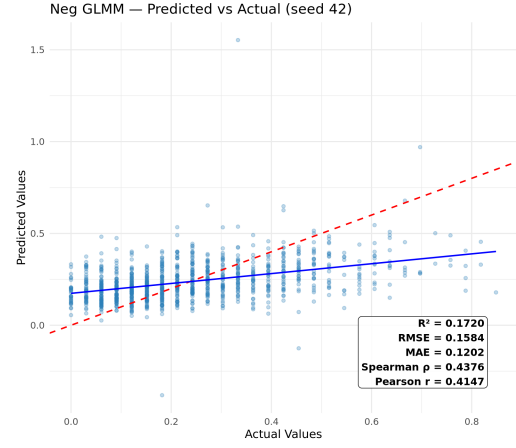
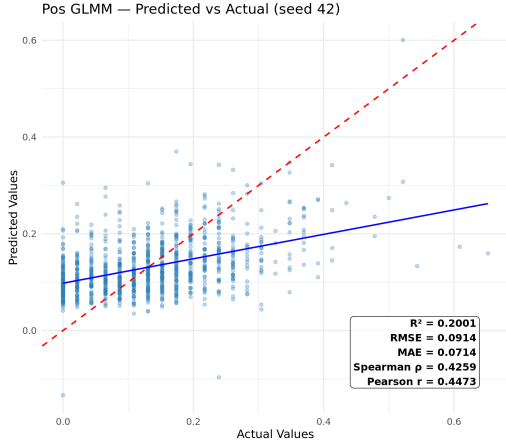


Figure 5.7: Calibration plots for GLMM models (positive and negative outcomes seed=42)

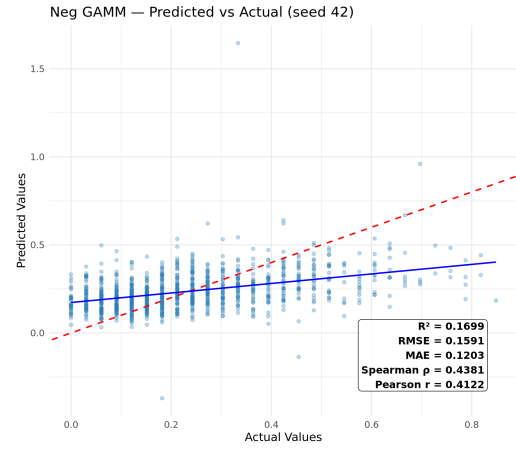
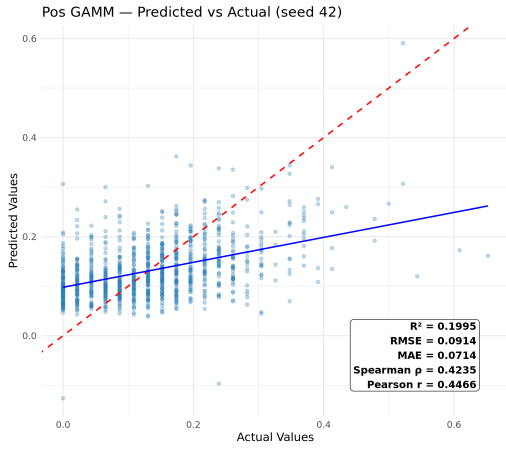


Figure 5.8: Calibration plots for GAMM models (positive and negative outcomes seed=42)

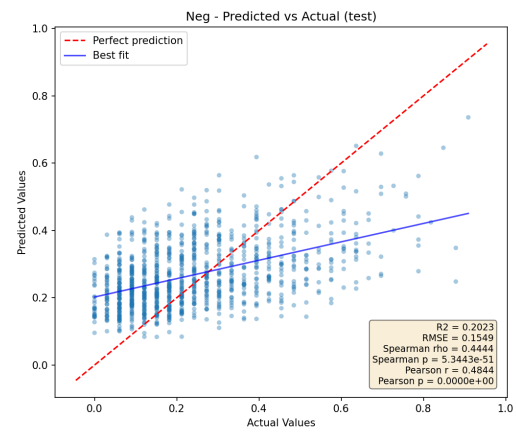
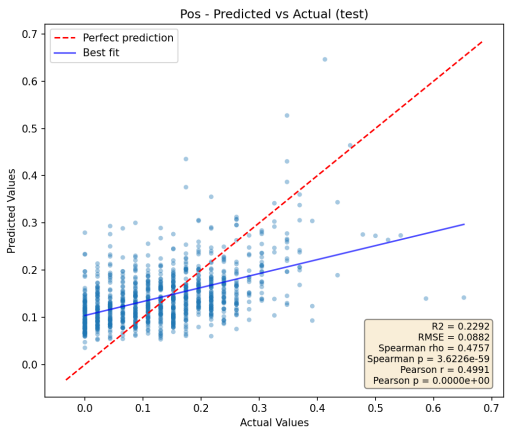


Figure 5.9: Calibration plots for DCMFNet models (positive and negative outcomes seed=42)

Overall, these results indicate that while statistical models provide interpretable and stable baselines, the deep learning approach offers improved flexibility and predictive capacity, particularly in identifying higher-risk individuals.

This aligns with the central research objective, indicating that explicitly modeling complex, non-linear interactions between genetic, environmental, and behavioral factors yields improved representation of schizophrenia risk, particularly in the upper tail of the outcome distribution.

To assess robustness, model performance was averaged across multiple random seeds (42, 43, 44, 45, 46) as shown in Table 5.1. The results show consistent trends across runs, with DCMFNet outperforming both GLMM and GAMM in terms of explained variance, prediction error, and rank-based correlation.

Notably, GLMM and GAMM exhibit nearly identical performance across all metrics, indicating that the additional flexibility introduced through smooth terms does not materially improve predictive accuracy in this setting. In contrast, DCMFNet provides consistent gains, particularly in Spearman correlation, suggesting an improved ability to capture the ordering of risk. Improvements in R^2 remain modest, indicating that the primary advantage of DCMFNet lies in its ability to model complex relationships in the data rather than in substantially increasing explained variance.

The relatively low variance across seeds further indicates that these differences are stable and not driven by data partitioning.

Model	R^2	RMSE	MAE	Spearman ρ
GLMM (Pos)	0.181 ± 0.020	0.094 ± 0.002	0.073 ± 0.002	0.422 ± 0.021
GAMM (Pos)	0.180 ± 0.020	0.094 ± 0.002	0.073 ± 0.002	0.420 ± 0.021
DCMFNet (Pos)	0.206 ± 0.019	0.090 ± 0.001	0.071 ± 0.000	0.450 ± 0.019
GLMM (Neg)	0.179 ± 0.021	0.161 ± 0.005	0.124 ± 0.003	0.435 ± 0.013
GAMM (Neg)	0.178 ± 0.021	0.162 ± 0.005	0.124 ± 0.003	0.434 ± 0.010
DCMFNet (Neg)	0.198 ± 0.021	0.156 ± 0.002	0.124 ± 0.002	0.451 ± 0.013

Table 5.1: Model performance averaged across multiple random seeds (mean $\pm$ standard deviation).

Chapter 6

Analysis and Interpretation

This section moves beyond predictive performance and examines how different models capture relationships between substance use, genetic risk, and other covariates. We focus on interpretability, interaction structure, and model-specific behavior.

6.1 GLMM

The main effects reveal that much of the predictive signal is diffuse rather than dominated by a few large coefficients as seen in Figure 6.1. Many baseline symptom measures at age 15 contribute small but consistently positive effects, suggesting that early behavioral signals act cumulatively rather than individually. This aligns with a “distributed risk” structure, where no single variable is decisive, but multiple weak signals jointly shape the outcome.

In contrast, genetic effects (PRS components) appear unstable in direction and magnitude across outcomes. This suggests that, within a linear framework, PRS may not act independently but instead interact with environmental exposures—an effect that is only partially captured in the main effects specification.

Notably, the negative coefficient observed for opioid-related substance use (*SUD15*) in the positive symptom model is difficult to interpret causally and likely reflects residual confounding or the dominance of interaction terms in capturing its contribution to risk. This highlights a limitation of GLMMs in high-dimensional settings, where correlated exposures can induce suppression effects.

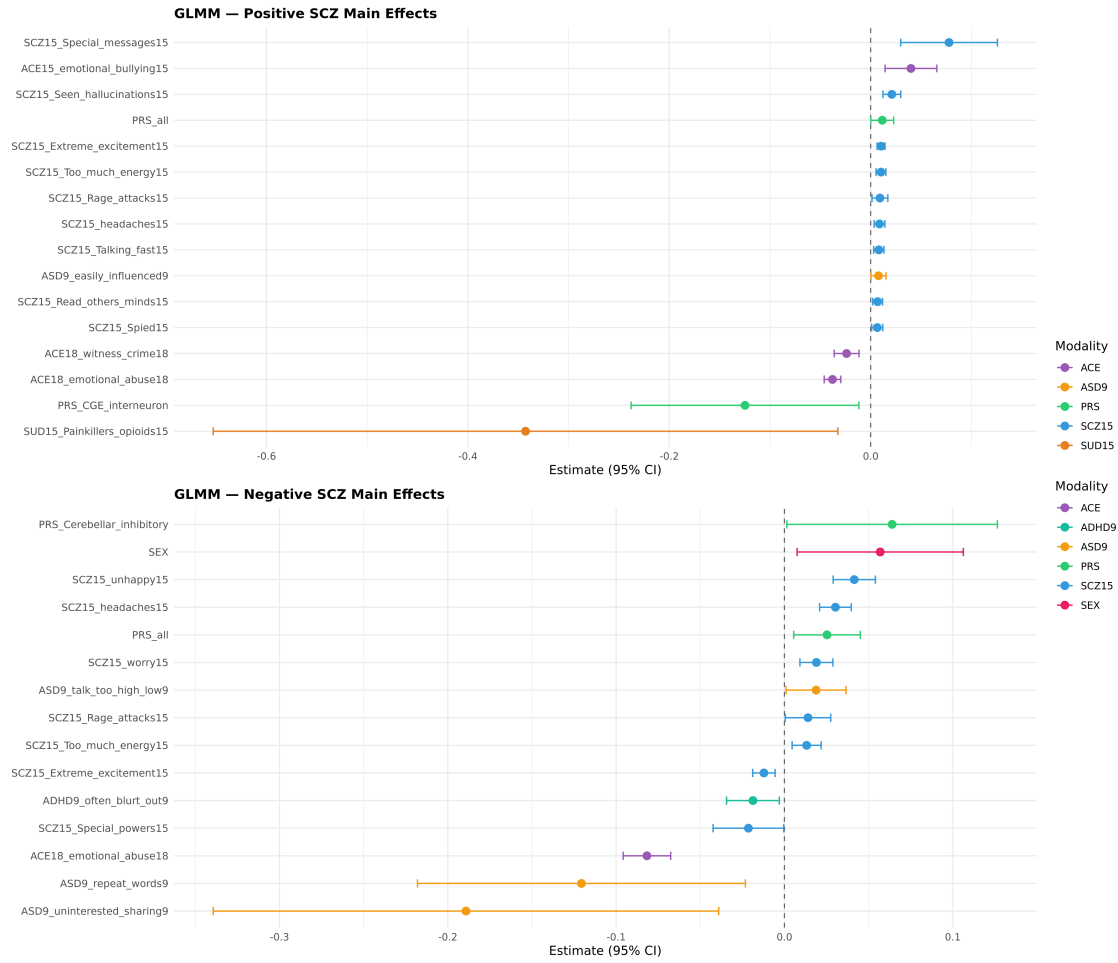


Figure 6.1: Main effects in GLMM models for positive and negative outcomes at seed 42 using regression coefficients.

The interaction structure (Figure 6.2) reveals more specific and non-uniform patterns. Interactions involving *other drugs* show consistent positive effects with PRS components (e.g., interneuron-related scores), suggesting a gene–environment coupling where genetic vulnerability is amplified under substance exposure. In contrast, *opioid use* exhibits a particularly strong positive interaction with severe adverse experiences (e.g., *ACE18_other_abuse18*), indicating that risk is sharply elevated only in the presence of both high-risk environments and specific substance use.

A different pattern emerges for *snuff* and some behavioral traits (e.g., ASD-related features), where interactions are negative or near zero. This suggests that not all substance use acts uniformly as a risk amplifier and may reflect heterogeneous behavioral subgroups or residual confounding.

For negative symptoms, interaction effects are generally weaker and more dispersed. The absence of strong, concentrated interactions suggests that negative symptom severity is less driven by acute exposure combinations and more by background or trait-like factors.

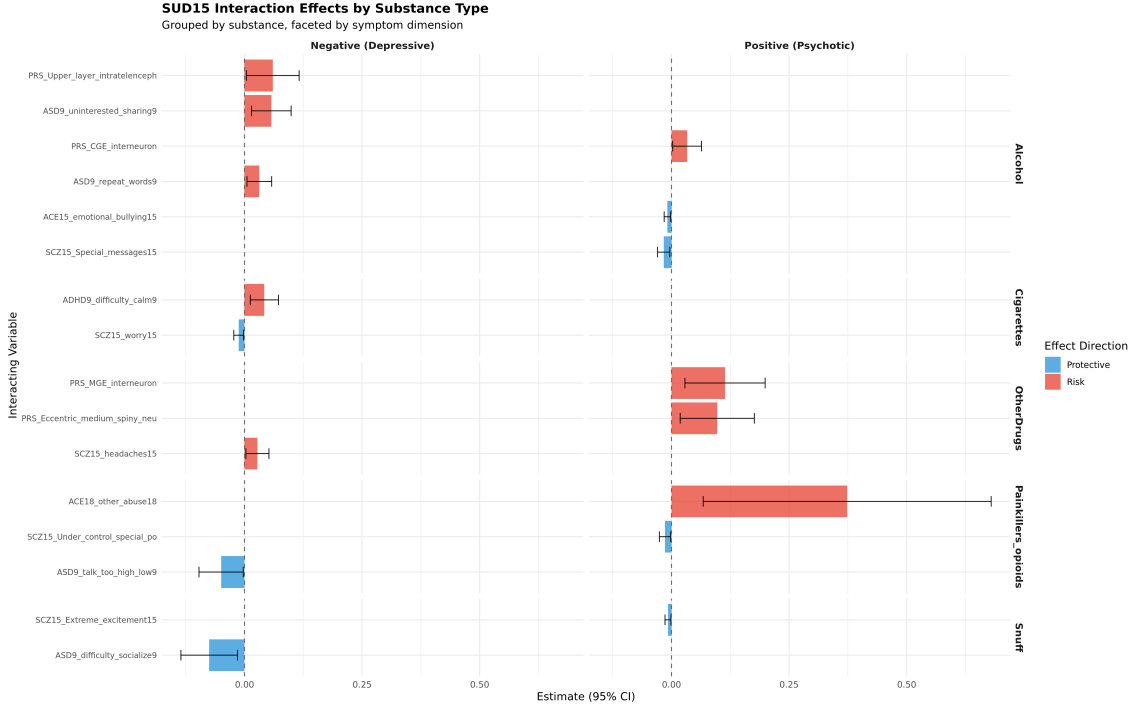


Figure 6.2: Interaction effects between substance use and covariates in GLMM models, stratified by substance type at seed 42 using regression coefficients.

Overall, the interaction plot indicates that risk is highly conditional: substance use amplifies vulnerability selectively, depending on the type of substance and the co-occurring environmental or genetic context.

6.2 GAMM

The GAMM results (Figure 6.3) are largely consistent with the GLMM, indicating that introducing smooth terms does not materially alter the main effect structure. In particular, the contribution of PRS remains similar in both direction and magnitude, suggesting limited non-linearity in its relationship with the outcomes.

The interaction patterns (Figure 6.4) also closely replicate those observed in the GLMM. The strong positive interaction between opioid use and severe adverse experiences (e.g., *ACE18_other_abuse18*) persists, as do the positive associations between *other drugs* and PRS components. These effects remain the dominant interaction signals in the model.

Only modest differences are observed between the GLMM and GAMM interaction structures. In general, interaction estimates appear slightly attenuated and more centralized around zero, reflecting the regularizing effect of the smooth terms applied to selected covariates.

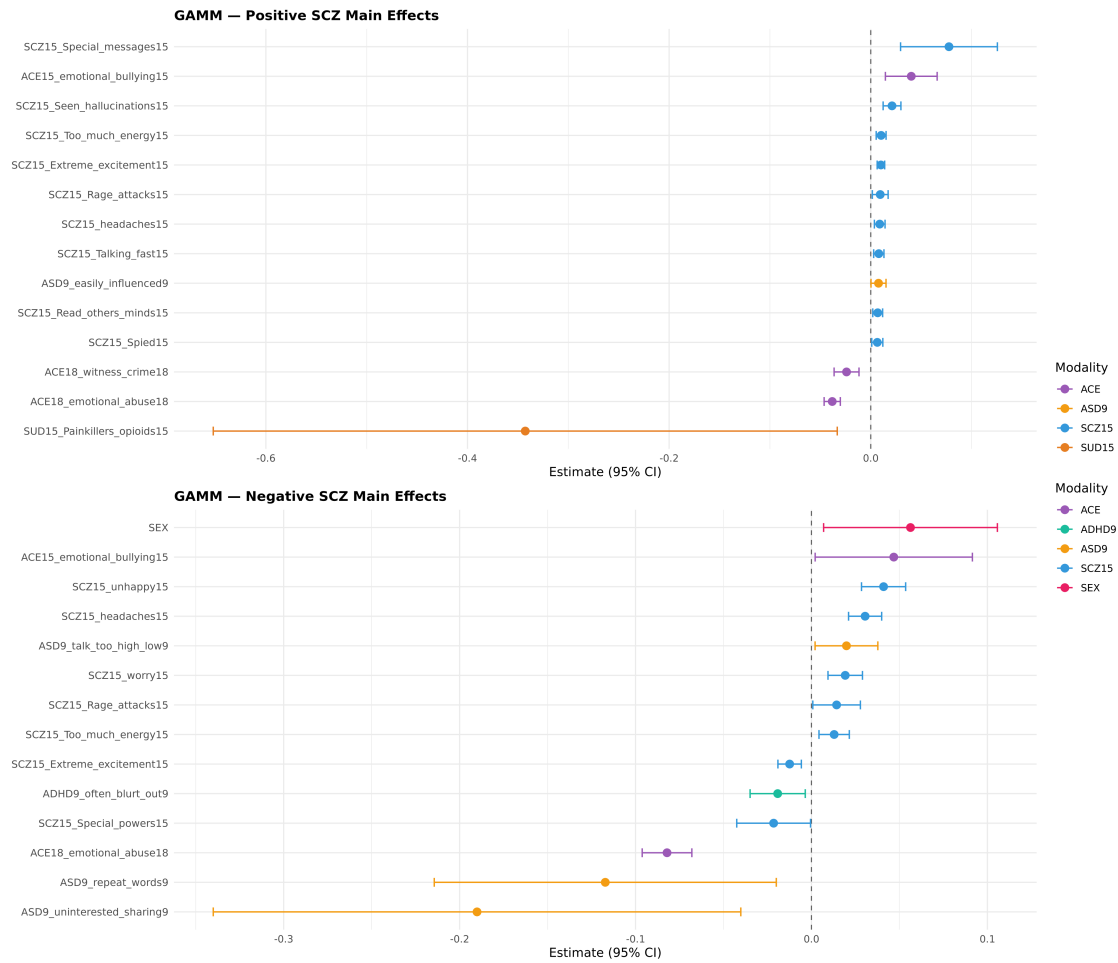


Figure 6.3: Main effects in GAMM models for positive and negative outcomes at seed 42 using regression coefficients.

In particular, interactions involving PRS and substance use (especially within the *OtherDrugs* category) remain consistently positive but exhibit reduced variability and tighter confidence intervals, suggesting that GAMM stabilizes these gene–environment effects rather than altering them. Similarly, the strong interaction between opioid use and severe adverse experiences (e.g., *ACE18_other_abuse18*) persists as the dominant signal, although its magnitude is marginally reduced, indicating that this effect is robust to model specification rather than an artifact of linearity.

In contrast, weaker interactions—such as those involving cigarette use and certain SCZ15 behavioral traits—are further shrunk toward zero and become less prominent, implying that these effects may reflect noise or model sensitivity in the GLMM framework. A similar attenuation is observed for negative (protective) interactions involving ASD-related traits and snuff use, which appear less pronounced under the GAMM specification.

For negative symptoms, interactions involving PRS components show slightly more consistent positive contributions, but remain relatively small in magnitude, reinforcing the view that negative symptom variation is less driven by strong exposure interactions.

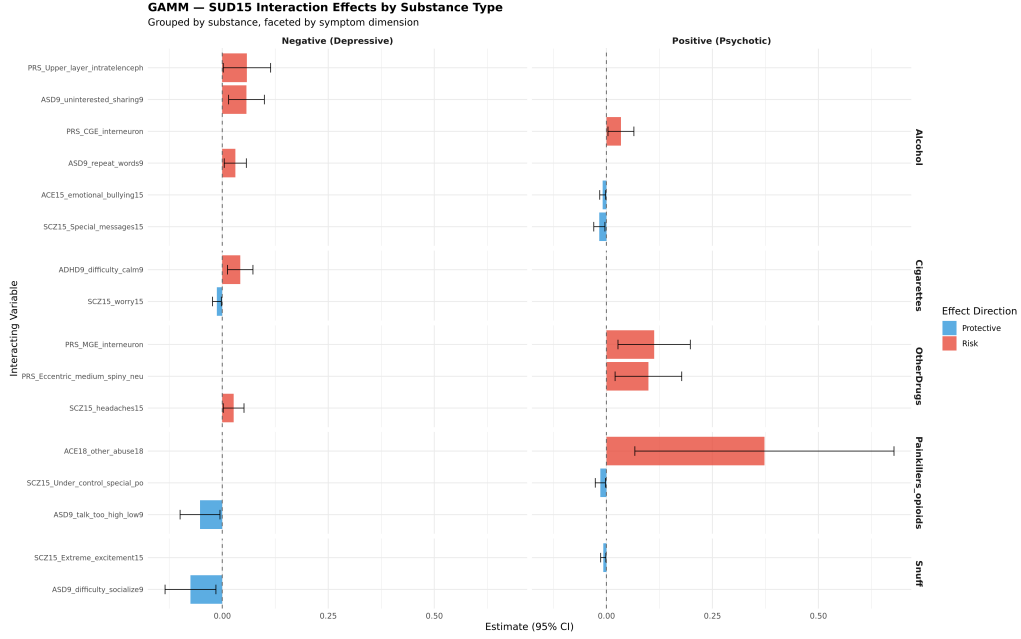


Figure 6.4: Interaction effects between substance use and covariates in GAMM models at seed 42 using regression coefficients.

Overall, GAMM does not introduce new high-impact interactions but instead refines the existing structure by dampening weaker effects and stabilizing dominant ones.

6.3 DCMFNet

6.3.1 Complexity Analysis

To better understand how interactions are represented within the DCMFNet architecture, we performed a complexity analysis by varying the depth of individual IGF blocks from 1 to 5 while holding others fixed at their default values. This optimal depth sweep was evaluated by minimizing RMSE.

The results (Figures 6.5 and 6.6) reveal clear heterogeneity in the required interaction complexity across modalities.

For the positive symptom model, several modalities benefit from deeper interaction structures. In particular, PRS, ASD-related features, and SEX reach optimal performance at higher depths (4–5 layers), indicating that their relationship with substance use is more compositional and likely involves higher-order interactions. In contrast, ADHD, ACE18, SES, and SUD18 achieve optimal performance at shallow depths (1–2 layers), suggesting more direct relationships. This pattern supports the idea that positive symptoms are driven by a mix of simple and highly complex interactions, requiring flexible modelling capacity.

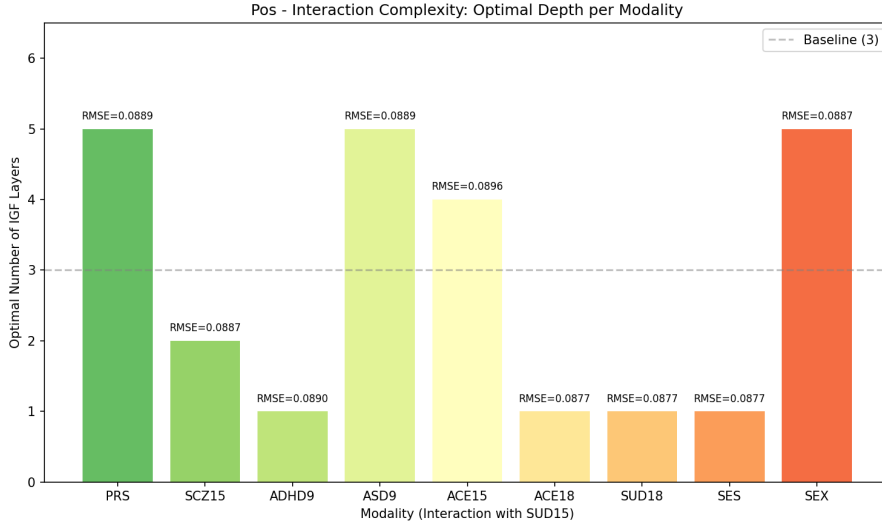


Figure 6.5: Optimal depth search for each modal block in the positive DCMFNet model at seed 42.

For the negative symptom model, however, the pattern is notably different. Most modalities achieve optimal performance at shallow depths (1–2 layers), and importantly, the overall performance does not improve meaningfully over the default configuration. In fact, the comparison between optimized and default settings indicates that the default depth already provides the best or near-best performance, suggesting that increasing interaction complexity does not yield additional predictive benefit for negative symptoms. This aligns with earlier findings from GLMM and GAMM, where negative symptoms showed weaker and more diffuse interaction effects.

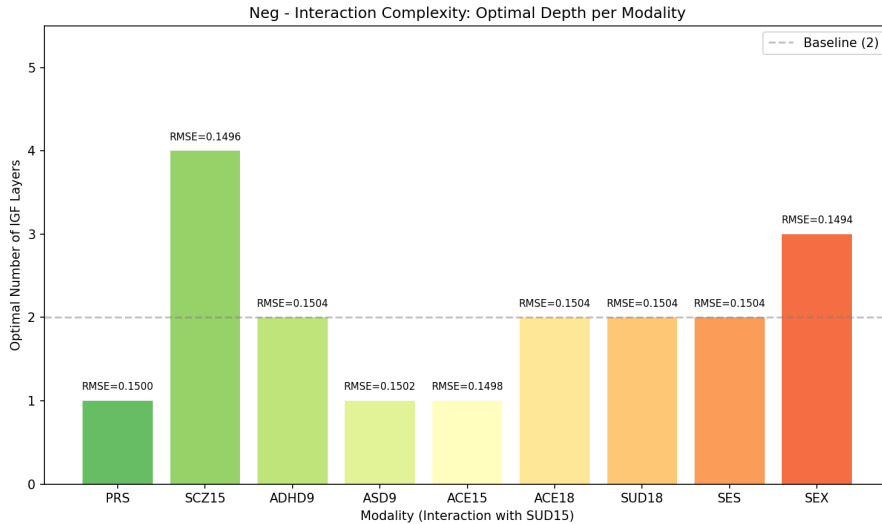


Figure 6.6: Optimal depth search for each modal block in the negative DCMFNet model at seed 42.

Although RMSE differences across depths are relatively small, the modality-specific

patterns are consistent—particularly for positive symptoms—indicating that the variation reflects underlying structure rather than noise. Adaptive interaction depth improves performance for positive symptoms, while offering little benefit for negative symptoms, where simpler interaction structures are sufficient.

In addition, an alternative strategy was explored where the number of layers in each IGF block was jointly optimized as part of the hyperparameter tuning process. While this approach allowed for a more flexible search over interaction complexity, the resulting models consistently exhibited overfitting when evaluated on the test set. This suggests that jointly optimizing a high-dimensional hyperparameter space—including per-block depths—may be unstable given the sample size and complexity of the data.

As a result, we retained the tuned hyperparameters obtained under a fixed-depth configuration and instead used the present complexity analysis to assess and select modality-specific interaction depths in a more controlled manner.

Importantly, this analysis provides a level of granularity not available in GLMM or GAMM. Rather than imposing a uniform interaction structure, DCMFNet reveals that different modalities operate at different levels of complexity. In particular, it highlights that positive symptoms are driven by heterogeneous, higher-order interactions, whereas negative symptoms are governed by more direct and shallow relationships—an insight that traditional models are not able to explicitly capture.

Model	Configuration	RMSE	R^2	Spearman ρ
Positive	Default (L=3)	0.0891	0.2129	0.4631
Positive	Optimal (Per-modality)	0.0863	0.2618	0.4988
Negative	Default (L=2)	0.1531	0.2209	0.4488
Negative	Optimal (Per-modality)	0.1551	0.2008	0.4354

Table 6.1: Comparison between default and optimized interaction depth configurations for DCMFNet models (test set performance) at seed 42.

A quantitative comparison between default and optimized configurations is provided in Table 6.1, confirming that adaptive depth improves performance for positive symptoms but not for negative symptoms.

Based on these results, the best-performing configuration was selected for subsequent analysis.

6.3.2 Importance Analysis

To interpret the DCMFNet model, feature importance was assessed using attention weights from both independent and fused representations.

Feature Importance Across the Severity Spectrum

The attention heatmaps (Figures 6.7 and 6.8) show that feature importance varies systematically across the schizophrenia severity spectrum, rather than remaining uniform. Each row represents a feature and each column an individual (ordered by increasing schizophrenia severity), with colour intensity indicating attention weight—warmer colours denote higher importance and cooler colours lower importance.

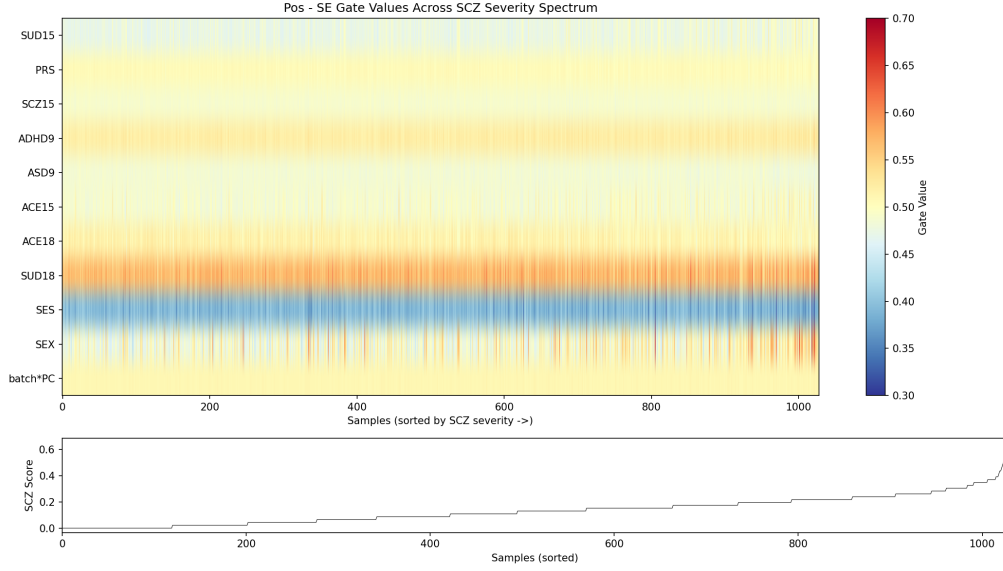


Figure 6.7: Feature importance for independent features in the positive DCMFNet model at seed 42.

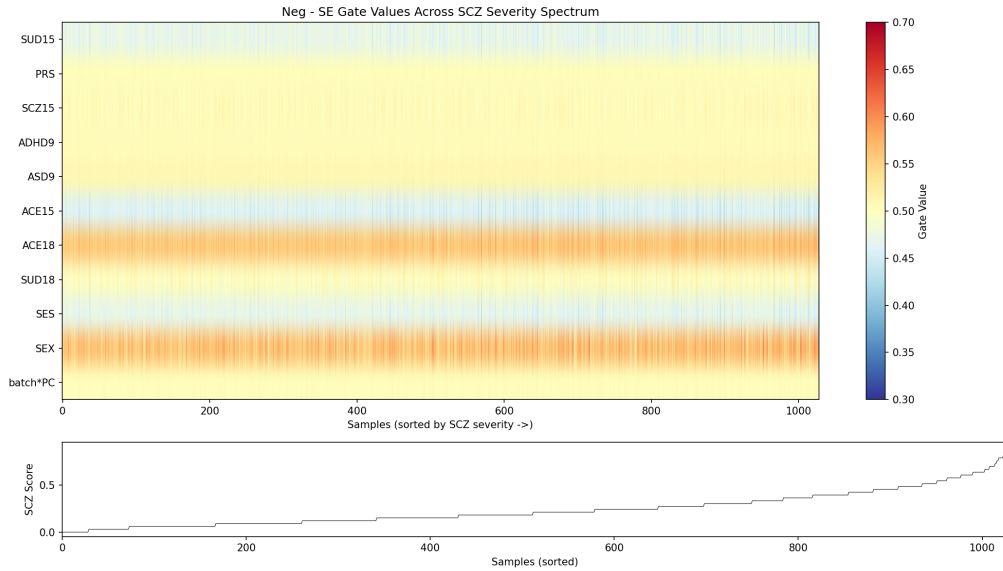


Figure 6.8: Feature importance for independent features in the negative DCMFNet model at seed 42.

For the positive model, importance shows a clear gradient with increasing severity. *SUD18* and *SEX* consistently receive high weights, with slight amplification in higher-risk individuals, while *ADHD9* also stands out with elevated importance. In contrast, *SES* remains consistently low, indicating limited direct contribution. Genetic (*PRS*) and baseline symptom (*SCZ15*) features exhibit stable, moderate importance, suggesting they provide a background signal rather than driving extremes.

For the negative model, importance is more uniform across individuals. Features such as *ACE18*, *SEX*, and *SUD18* maintain consistently high weights with minimal variation across severity, indicating a more stable, trait-like structure. Compared to the positive model, there is little evidence of dynamic reweighting.

Overall, the heatmaps indicate that DCMFNet adaptively reweights features across the severity spectrum, with stronger heterogeneity in the positive model and more uniform contributions in the negative model.

Feature importance derived from DCMFNet attention gates is not directly comparable to effect sizes in GLMM/GAMM or importance measures from tree-based models such as LightGBM.

In statistical models, importance reflects the magnitude and significance of explicitly specified associations, while in tree-based models it reflects contribution to prediction error through splits.

In contrast, the SE attention gates in DCMFNet scale feature representations within the model. The extracted gate values therefore reflect how strongly each modality is weighted during prediction, rather than its marginal effect or predictive gain.

As a result, features like *SUD18* may receive high attention weights without appearing among the top predictors in GLMM or GAMM. This indicates that the model relies on these features within its internal representation, even if their isolated statistical effects are weak or unstable.

Attention-based importance should therefore be interpreted as model reliance within a learned representation, providing complementary insight into how information is distributed across features during prediction.

High- vs Low-Risk Group Comparison

To further characterize this behavior, individuals were stratified into high-risk (top 15%) and low-risk groups. Differences in attention weights were assessed using the Mann–Whitney U test, a non-parametric test suitable for comparing distributions without assuming normality. In these plots, each bar represents the average attention weight for a feature within a group, with error bars indicating variability. Differences between high- and low-risk groups are reflected by differences in bar height, while asterisks denote statistically significant differences.

At the interaction (fused) level, differences between groups are minimal (Figure 6.9), indicating that the global interaction structure is largely shared across individuals.

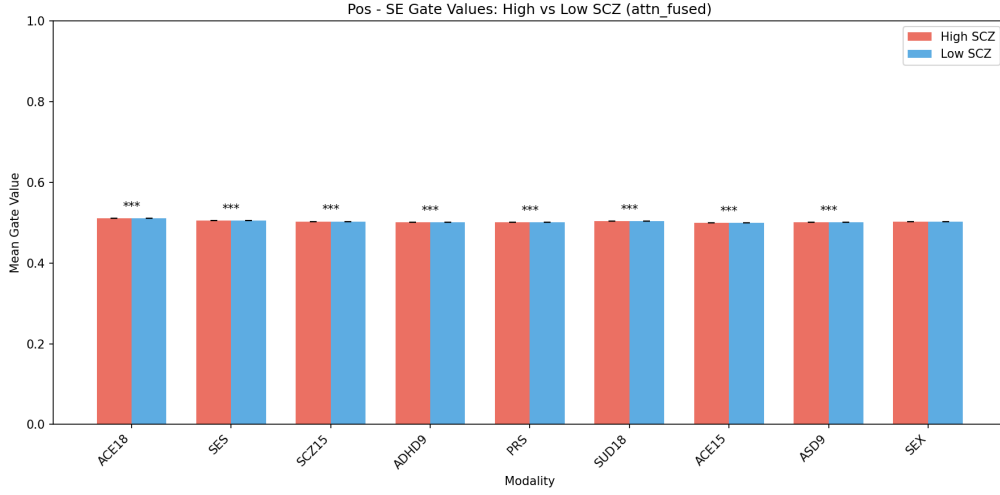


Figure 6.9: Interaction feature importance comparison between high- and low-risk groups (positive model) at seed 42.

In contrast, independent features show clearer differentiation (Figure 6.10). High-risk individuals place greater emphasis on *SEX*, *SUD15*, *ADHD9*, and *ACE15*, highlighting the role of early behavioral and environmental factors in driving elevated risk. *SUD18* remains highly weighted overall but does not strongly differentiate between groups, suggesting a general rather than discriminative effect. Features such as *SES* and *batch*PC* contribute weakly to stratification, while *PRS* shows modest but consistent shifts.

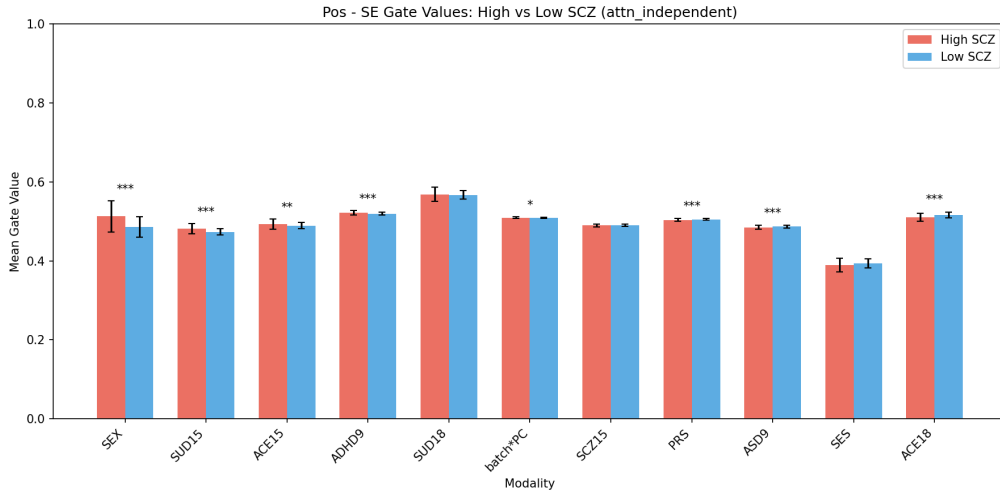


Figure 6.10: Independent feature importance comparison between high- and low-risk groups (positive model) at seed 42.

For the negative model, both fused and independent representations show only small and uniform differences between groups (Figures 6.11 and 6.12), indicating limited subgroup-specific structure. For the negative model, a further distinction emerges between representation levels. While fused (interaction) features show little to no statistically meaningful differences between high- and low-risk groups, several independent features remain sig-

nificant, albeit with small effect sizes. This suggests that, for negative symptoms, risk differentiation is driven primarily by additive contributions of individual features rather than higher-order interaction effects.

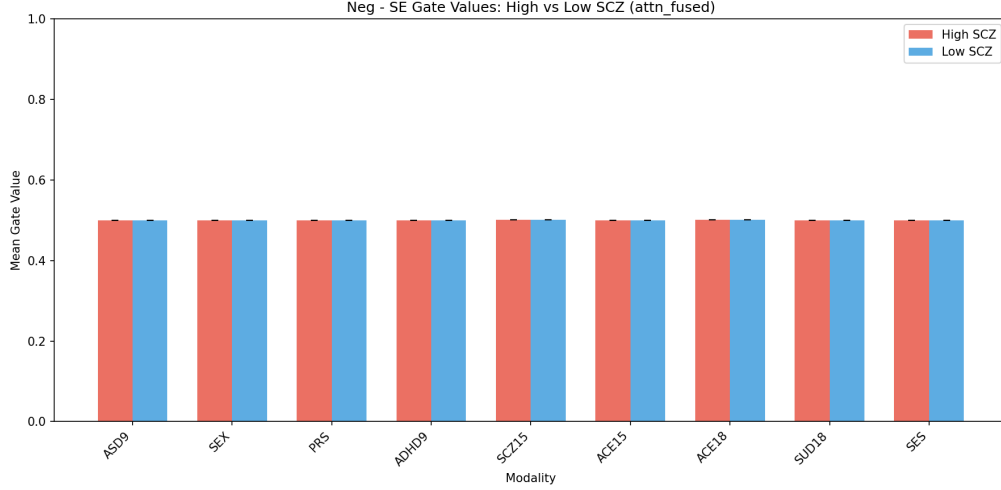


Figure 6.11: Interaction feature importance comparison between high- and low-risk groups (negative model) at seed 42.

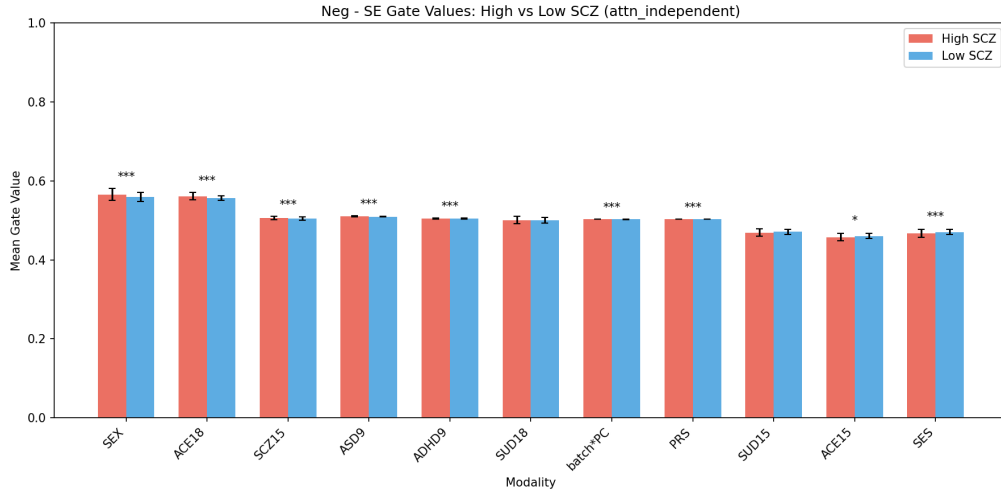


Figure 6.12: Independent feature importance comparison between high- and low-risk groups (negative model) at seed 42.

Overall, risk stratification is primarily driven by shifts in independent feature importance in the positive model, whereas the negative model exhibits more homogeneous patterns. This suggests that positive symptoms are influenced by context-dependent amplification of behavioral and environmental factors, while negative symptoms reflect more stable, trait-like mechanisms.

These findings align with developmental models of schizophrenia, where risk emerges from the interaction between genetic liability, environmental exposures, and behavioral

trajectories. In particular, the results suggest that while interaction structures are broadly shared across individuals, variation in feature weighting determines progression toward higher risk.

6.4 ML – Benchmark Analysis

To provide additional context, we compared model performance with standard machine learning approaches, averaged across five random seeds.

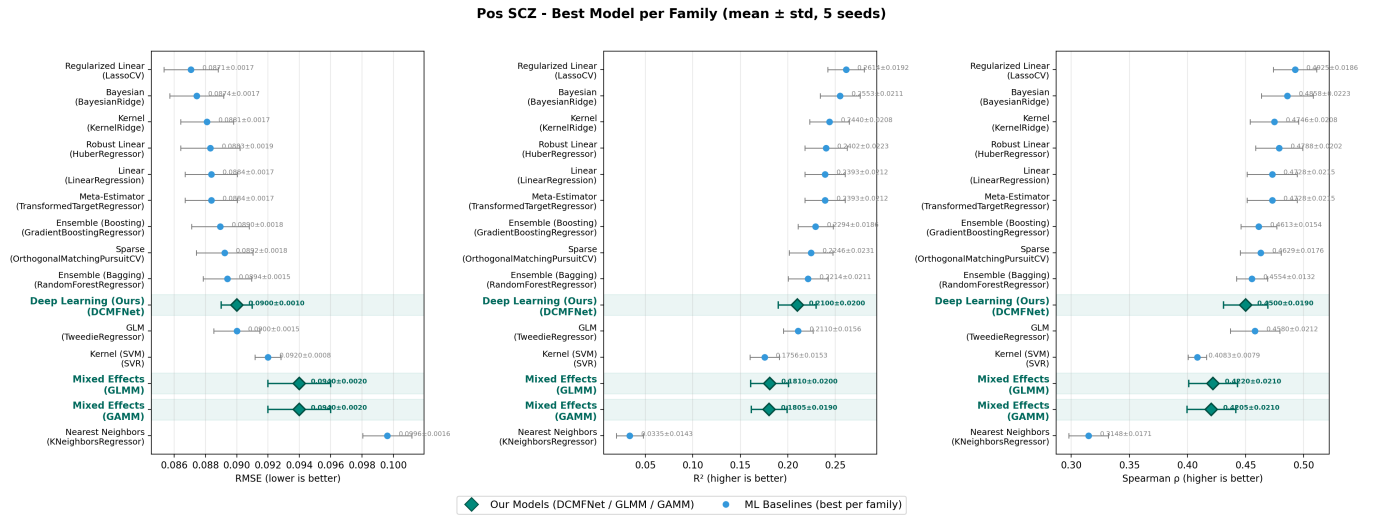


Figure 6.13: Model performance comparison by algorithm family for positive outcomes.

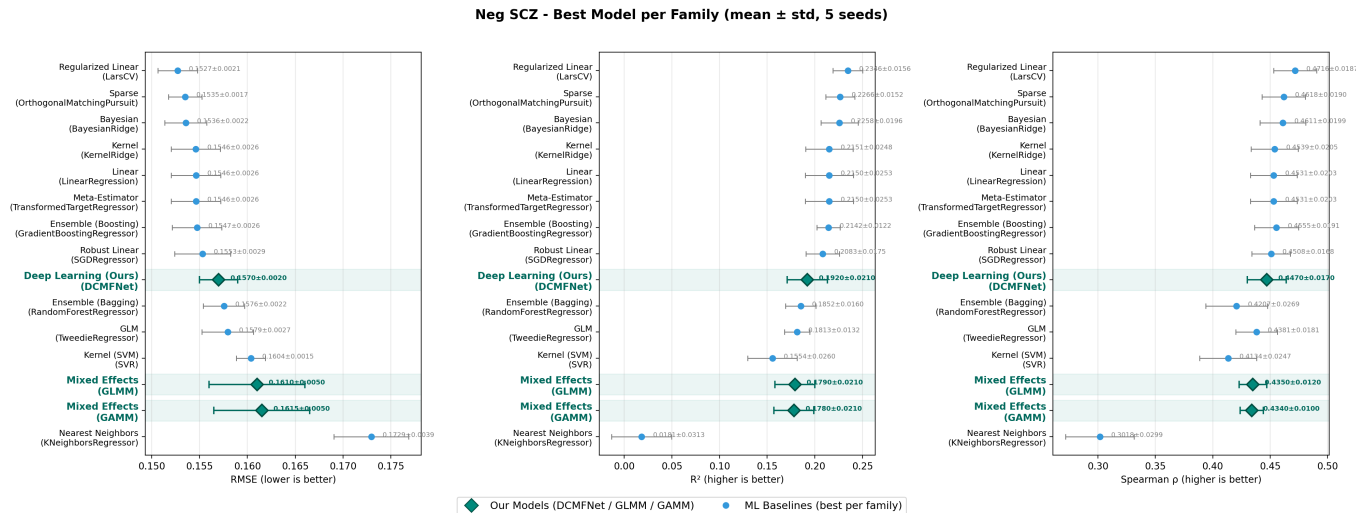


Figure 6.14: Model performance comparison by algorithm family for negative outcomes.

A consistent pattern across both positive and negative outcomes is that DCMFNet outperforms the structured statistical models (GLMM and GAMM) across all evaluation metrics. In addition to improved mean performance, DCMFNet also exhibits lower variance across seeds, indicating more stable and robust behavior.

In contrast, both GLMM and GAMM show wider standard deviations, suggesting greater sensitivity to data splits and reduced stability in estimation.

When compared to standard machine learning models, performance differences are more nuanced. Several ML models (e.g., regularized linear models and kernel methods) achieve comparable or slightly higher performance in certain metrics. However, these gains are generally small and often fall within overlapping uncertainty ranges.

A clearer distinction emerges between the two outcomes. For negative symptoms, model performance is tightly clustered across all methods, suggesting that the underlying structure is relatively simple and does not strongly benefit from increased model flexibility. In contrast, the positive outcome exhibits greater spread across models, indicating a more complex signal where modeling choices have a larger impact.

Notably, instance-based methods such as K-nearest neighbors perform poorly and show high variability, particularly for the negative outcome. This suggests that local similarity-based approaches are not well suited to the structured and high-dimensional nature of the data.

Overall, these results highlight a key trade-off. While several established machine learning models achieve stronger predictive performance, DCMFNet provides a useful balance between accuracy, stability, and interpretability. In particular, its consistent improvement over GLMM and GAMM suggests that explicitly modeling interaction structure offers advantages over predefined or smoothly constrained approaches. These results represent an initial step toward incorporating such architectures in this domain, rather than a replacement for well-established machine learning methods.

Chapter 7

Discussion

7.1 Main Findings

This study examined how adolescent substance use interacts with genetic, environmental, and behavioral factors to influence psychotic symptoms, using both structured statistical models (GLMM, GAMM) and a deep learning approach (DCMFNet). Three main findings emerge.

First, interactions involving substance use are central to understanding risk, particularly for positive symptoms. Across both GLMM and GAMM, the strongest effects consistently arise from interactions between substance use and ACE, as well as substance use and genetic risk (PRS). These effects are selective rather than uniform, indicating that substance use acts as a context-dependent amplifier rather than an independent risk factor.

Second, positive and negative symptoms exhibit fundamentally different structures. Positive symptoms are characterized by strong, targeted interaction effects and benefit from increased model complexity. In contrast, negative symptoms show weaker and more diffuse interaction patterns, with greater influence from baseline symptoms and confounding structure, and limited gains from more complex modeling.

Third, the choice of modeling framework affects both performance and stability.

DCMFNet consistently outperforms GLMM and GAMM across metrics, while also exhibiting lower variability across random seeds. Importantly, its advantage is most visible in the upper range of the outcome distribution, where it better distinguishes higher-risk individuals. In contrast, GLMM and GAMM tend to regress toward the mean and underestimate these cases.

Together, these findings suggest that schizophrenia risk is driven by structured interactions that differ across symptom domains, and that modeling flexibility is particularly important for capturing high-risk individuals.

7.2 Connection to Existing Literature

These findings support the view of schizophrenia as a multi-factorial disorder driven by gene–environment interactions, while refining how these effects operate. In particular, cell-type specific PRS emerge as important contributors, highlighting that genetic risk is not uniform but operates through biologically distinct pathways. Interactions between substance use and these cell-type PRS suggest that genetic vulnerability may be differentially amplified depending on the underlying cellular mechanisms.

Consistent with prior work, substance use does not act independently but amplifies risk in the presence of adverse experiences and genetic liability. However, this amplification is selective rather than uniform, varying across substances and contexts. ADHD- and ASD-related traits also emerge as important contributors, aligning with literature linking early neuro-developmental features to later psychosis risk, and reinforcing their role within broader interaction structures rather than as isolated predictors.

7.3 Implications for Mental Health Research

The implications of these findings are primarily clinical rather than methodological.

First, risk estimates should be interpreted as conditional rather than absolute. For an individual, factors such as substance use do not carry uniform risk, but depend on their co-occurrence with adverse experiences, baseline symptoms, and genetic vulnerability. This suggests that clinical assessment should move away from single-factor screening toward evaluating combinations of risk factors within a broader context.

Second, the ability to better identify individuals at the higher end of the risk spectrum is particularly relevant. Across models, predictions tend to regress toward the mean, but DCMFNet shows improved sensitivity to higher-risk cases. In practice, this translates to a greater ability to flag individuals who may benefit from early monitoring or intervention, even if overall predictive performance differences appear modest.

Third, the differing structure of positive and negative symptoms implies that they may require distinct clinical strategies. Positive symptoms appear more sensitive to environmental and behavioral interactions, suggesting potential value in early behavioral or exposure-based interventions. In contrast, negative symptoms, being more stable and less interaction-driven, may require approaches focused on long-term support rather than short-term risk modification.

Overall, these results suggest that predictive models in mental health should prioritize identifying high-risk individuals and capturing context-dependent risk, rather than optimizing average prediction accuracy alone.

7.4 Strengths and Limitations

Each modeling approach contributes complementary strengths. GLMM provides a clear and interpretable framework for estimating structured effects and accounting for dependencies such as the twin design. GAMM extends this by allowing flexible, non-linear relationships while retaining interpretability and statistical inference. DCMFNet, in contrast, offers greater flexibility in modeling interaction structure, leading to improved predictive performance and stability, particularly in capturing higher-risk individuals. Importantly, the different modeling approaches do not provide competing explanations, but rather complementary perspectives on the same underlying system, each capturing different aspects of the relationship between predictors and outcome.

The main limitation of this study is data-related rather than methodological, specifically the effective sample size relative to model complexity. Although the dataset is large in absolute terms, high dimensionality, missingness, and stratification reduce the information available for learning complex interaction structures. This is reflected in the modest R^2 values across all models, indicating that a substantial proportion of variance remains unexplained.

Additionally, while overall trends remained consistent across seeds, model performance and learned representations were sensitive to random seed, particularly for DCMFNet, indicating instability in the optimization landscape under limited effective sample size.

7.5 Ethics

This study was conducted in accordance with the TRIPOD+AI guidelines [20], ensuring transparent reporting and evaluation of predictive models. The use of observational data requires careful interpretation, particularly when modeling sensitive outcomes such as mental health risk.

All analyses were performed on de-identified data, and no attempt was made to derive individual-level predictions for clinical use. The models are intended to support understanding of population-level patterns rather than decision-making at the individual level.

Special care was taken to avoid deterministic interpretations of genetic risk. PRS variables reflect probabilistic associations and should not be interpreted as fixed or predictive at the individual level. Similarly, associations with behavioral or environmental exposures do not imply causation or inevitability.

7.6 Declaration of AI

Artificial intelligence tools were used to assist in code development and refinement of written text. All outputs were critically reviewed and adapted by the author. The ideas, interpretations, and conclusions presented in this work are the author’s own.

Chapter 8

Conclusions and Future Work

8.1 Conclusions

This work set out to model schizophrenia risk through the lens of interactions, but the more important outcome is how it re-frames the problem itself.

The results suggest that risk is not something that can be attributed to individual factors in isolation, nor fully captured by fixed interaction terms. Instead, it reflects a structured and context-dependent process, where a small number of dominant factors—such as adverse experiences and prior symptoms—form a baseline, and other exposures, including substance use, modify this risk in selective ways.

What emerges is not a single model that best explains the data, but a hierarchy of perspectives. GLMM and GAMM make the structure visible and interpretable, revealing which factors and interactions matter. DCMFNet builds on this by representing these relationships more flexibly, improving stability and sensitivity to higher-risk individuals. The value, therefore, lies not in replacing one approach with another, but in combining interpretability with flexibility.

More broadly, this study highlights a shift in emphasis: from identifying risk factors to understanding how they combine. In this setting, modeling choices are not just technical decisions, but they determine what aspects of risk can be observed.

Taken together, the findings suggest that progress in modeling schizophrenia risk will depend less on finding new predictors, and more on better representing how known factors interact within an individual’s developmental context. In this sense, schizophrenia risk is not defined by single exposures, but by how genetic, environmental, and behavioral factors evolve and interact across development.

8.2 Future Work

Future work can build on this foundation in several directions. The internal behavior of DCMFNet can be further analyzed to better understand how interactions are formed across layers. Random effects from the mixed models can be explored to identify mean-

ingful subgroups or outliers. In addition, high-performing machine learning models can be systematically tuned to provide stronger benchmarks.

More broadly, this work shows that it is now feasible to model structured interactions between substance use and multiple domains. Future studies can extend DCMFNet to capture richer interaction patterns across features, enabling a more detailed understanding of how complex risk factors combine in schizophrenia.

Code Availability

Code is available at: <https://github.com/sil-pixel/Thesis>

Bibliography

- [1] Na Zhan et al. “The genetic basis of onset age in schizophrenia: evidence and models”. In: *Frontiers in Genetics* Volume 14 - 2023 (2023). ISSN: 1664-8021. DOI: 10.3389/fgene.2023.1163361. URL: <https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2023.1163361>.
- [2] Neale MC Sullivan PF Kendler KS. “Schizophrenia as a Complex Trait: Evidence From a Meta-analysis of Twin Studies”. In: *Arch Gen Psychiatry* (2003). DOI: doi: 10.1001/archpsyc.60.12.1187.
- [3] Enkhmaa Luvsannyam et al. “Neurobiology of Schizophrenia: A Comprehensive Review”. In: *Cureus* 14.4 (2022), e23959. DOI: 10.7759/cureus.23959. URL: <http://dx.doi.org/10.7759/cureus.23959>.
- [4] Shuyang Yao et al. “Connecting genomic results for psychiatric disorders to human brain cell types and regions reveals convergence with functional connectivity”. In: *Nature Communications* 16.1 (2025), p. 395. DOI: 10.1038/s41467-024-55611-1. URL: <https://doi.org/10.1038/s41467-024-55611-1>.
- [5] Natassia Robinson and Sarah E. Bergen. “Environmental Risk Factors for Schizophrenia and Bipolar Disorder and Their Relationship to Genetic Risk: Current Knowledge and Future Directions”. In: *Frontiers in Genetics* Volume 12 - 2021 (2021). ISSN: 1664-8021. DOI: 10.3389/fgene.2021.686666. URL: <https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2021.686666>.
- [6] Camila Maria Severi Martins et al. “Emotional Abuse in Childhood Is a Differential Factor for the Development of Depression in Adults”. In: *The Journal of Nervous and Mental Disease* 202.11 (2014).
- [7] Ciaran Shannon et al. “M31. THE ASSOCIATION BETWEEN BULLYING VICTIMIZATION AND PSYCHOSIS AND THE MEDIATING ROLE OF PEER AND FAMILY SOCIAL SUPPORT”. In: *Schizophrenia Bulletin* 46.Supplement₁ (May 2020), S146–S146.
- [8] Amandeep Jutla, Jennifer Foss-Feig, and Jeremy Veenstra-VanderWeele. “Autism spectrum disorder and schizophrenia: An updated conceptual review”. In: *Autism Research* 15.3 (2022), pp. 384–412. DOI: <https://doi.org/10.1002/aur.2659>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/aur.2659>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/aur.2659>.

- [9] S. Dalsgaard et al. “Association between Attention-Deficit Hyperactivity Disorder in childhood and schizophrenia later in adulthood”. In: *European Psychiatry* 29.4 (2014), pp. 259–263. DOI: 10.1016/j.eurpsy.2013.06.004.
- [10] Esben Agerbo et al. “Polygenic Risk Score, Parental Socioeconomic Status, Family History of Psychiatric Disorders, and the Risk for Schizophrenia: A Danish Population-Based Study and Meta-analysis”. In: *JAMA Psychiatry* 72.7 (July 2015), pp. 635–641. ISSN: 2168-622X. DOI: 10.1001/jamapsychiatry.2015.0346. eprint: <https://jamanetwork.com/journals/jamapsychiatry/articlepdf/2211892/yoi150019.pdf>. URL: <https://doi.org/10.1001/jamapsychiatry.2015.0346>.
- [11] Brooke Carter et al. “Sex and gender differences in symptoms of early psychosis: a systematic review and meta-analysis”. In: *Archives of Women’s Mental Health* 25.4 (2022), pp. 679–691.
- [12] Kimberly Siletti et al. “Transcriptomic diversity of cell types across the adult human brain”. In: *Science* 382.6667 (2023), eadd7046. DOI: 10.1126/science.add7046. eprint: <https://www.science.org/doi/pdf/10.1126/science.add7046>. URL: <https://www.science.org/doi/abs/10.1126/science.add7046>.
- [13] Ashley E. Tate et al. “Predicting mental health problems in adolescence using machine learning techniques”. In: *PLOS ONE* 15.4 (Apr. 2020), pp. 1–13. DOI: 10.1371/journal.pone.0230389. URL: <https://doi.org/10.1371/journal.pone.0230389>.
- [14] Zhen Huang et al. “DCMFNet: Deep Cross-Modal Fusion Network for Different Modalities with Iterative Gated Fusion”. In: *Association for Computing Machinery. GI ’24* (2024). DOI: 10.1145/3670947.3670956. URL: <https://doi.org/10.1145/3670947.3670956>.
- [15] Shankar Rao Pandala. *LazyPredict*. Version 0.3.0. Mar. 2026. URL: <https://github.com/shankarpandala/lazypredict>.
- [16] Tian Ge et al. “Polygenic prediction via Bayesian regression and continuous shrinkage priors”. In: *Nature Communications* 10.1 (2019), p. 1776. DOI: 10.1038/s41467-019-09718-5. URL: <https://doi.org/10.1038/s41467-019-09718-5>.
- [17] Han Wang. “The association between mechanism-informed polygenic risk score and schizophrenia and treatment resistance”. Master’s thesis. Stockholm, Sweden: Karolinska Institutet, 2025.
- [18] Zhi-Hua Zhou. “Ensemble Methods: Foundations and Algorithms (1st ed.)” In: *Chapman and Hall/CRC* (2012). DOI: <https://doi.org/10.1201/b12207>.
- [19] Jie Hu et al. “Squeeze-and-Excitation Networks”. In: (2019). arXiv: 1709.01507 [cs.CV]. URL: <https://arxiv.org/abs/1709.01507>.

- [20] Gary S Collins et al. “TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods”. In: *BMJ* 385 (2024). DOI: 10.1136/bmj-2023-078378. eprint: <https://www.bmj.com/content/385/bmj-2023-078378.full.pdf>. URL: <https://www.bmj.com/content/385/bmj-2023-078378>.

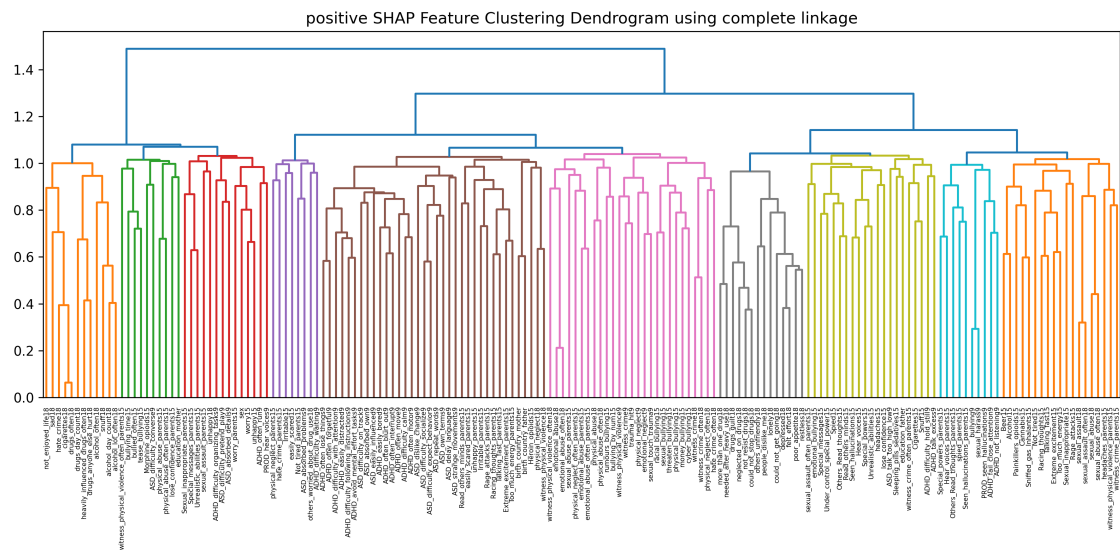


Figure A.2: SHAP Clustering of Positive LightGBM model using Complete linkage

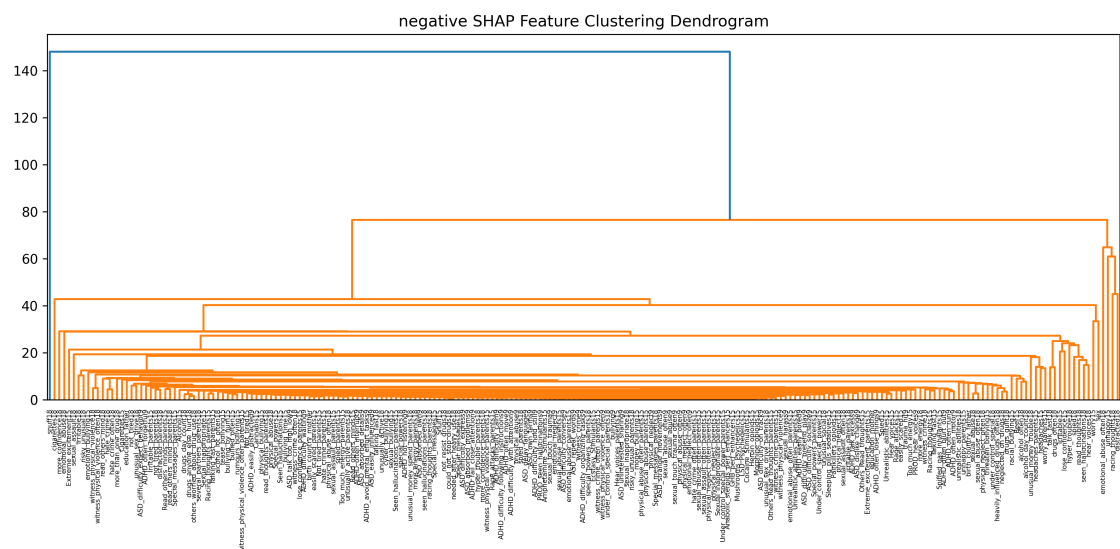


Figure A.3: SHAP Clustering of Negative LightGBM model using Ward linkage

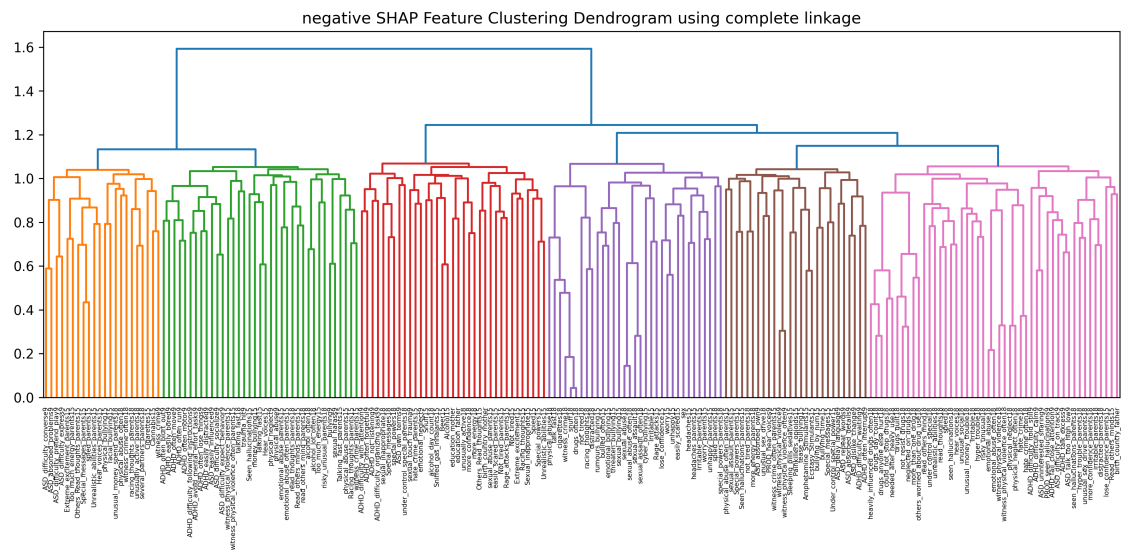


Figure A.4: SHAP Clustering of Negative LightGBM model using Complete linkage

Appendix B

DCMFNet Model Training

B.1 Hyperparameters

The tuned models exhibit lower weight decay, adjusted learning rates, and increased architectural flexibility, indicating a shift toward reduced regularization and greater capacity to capture complex interactions.

Hyperparameter	Default	Positive Model	Negative Model
Learning rate	1.82×10^{-4}	1.00×10^{-4}	3.22×10^{-4}
Batch size	8	4	6
Epochs	15	23	20
Optimiser	Adam	Adam	Adam
Weight decay	1.00×10^{-3}	2.95×10^{-5}	1.79×10^{-4}
Number of layers	5	3	2
Dropout	0.30	0.234	0.326
SE reduction	2	6	5
Min hidden dimension	8	13	5
Base loss	Huber	Huber	Huber
Huber δ	0.05	0.086	0.116
Focal γ	2.0	1.32	2.62
Number of bins	10	13	10
Scheduler patience	3	4	2
Scheduler factor	0.5	0.432	0.549
Early stopping patience	5	5	7
Seed	42	42	42

Table B.1: Comparison of default and tuned hyperparameters for DCMFNet models.

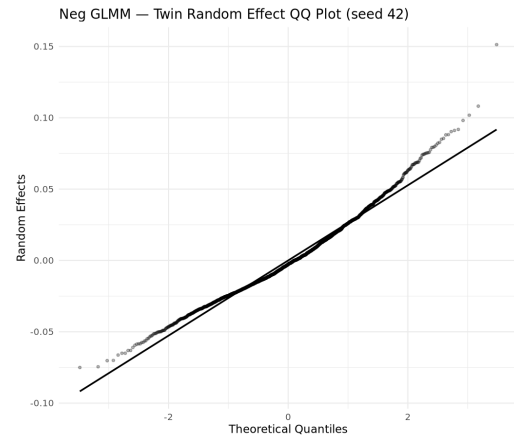
Appendix C

Results

C.1 Random Effects

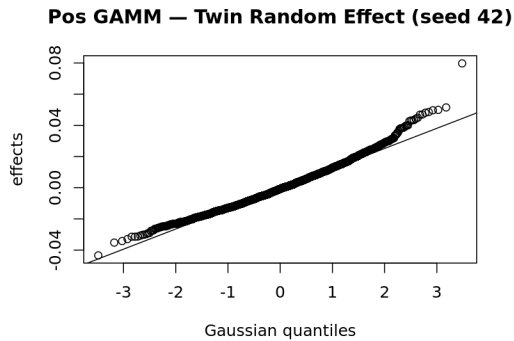


(a) Random effect of Twin in Positive GLMM Model at seed 42

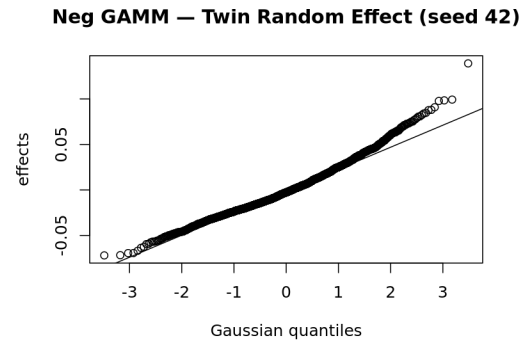


(b) Random effect of Twin in Negative GLMM Model at seed 42

Figure C.1: Random effect of Twin in GLMM Models at seed 42



(a) Random effect of Twin in Positive GAMM Model at seed 42



(b) Random effect of Twin in Negative GAMM Model at seed 42

Figure C.2: Random effect of Twin in GAMM Models at seed 42

QQ plots of the twin-level random effects indicate approximate normality for both GLMM and GAMM models as seen in Figures C.1 and C.2. While most points align closely with the theoretical quantiles, mild deviations are observed in the tails, particularly for higher quantiles, suggesting the presence of a small number of twin pairs with larger-than-expected effects.

These deviations are limited in magnitude and do not indicate substantial model misspecification. The Gaussian random intercept therefore provides a reasonable approximation for capturing within-pair dependence in this dataset.

The smooth terms for cell-type specific PRS (Figures C.3– C.4) show largely weak and near-linear effects, with most confidence intervals overlapping zero across the observed range. This suggests limited non-linear contribution of PRS components when modeled independently. Some components (e.g., interneuron-related scores) exhibit mild directional trends, but overall effects remain small and uncertain.

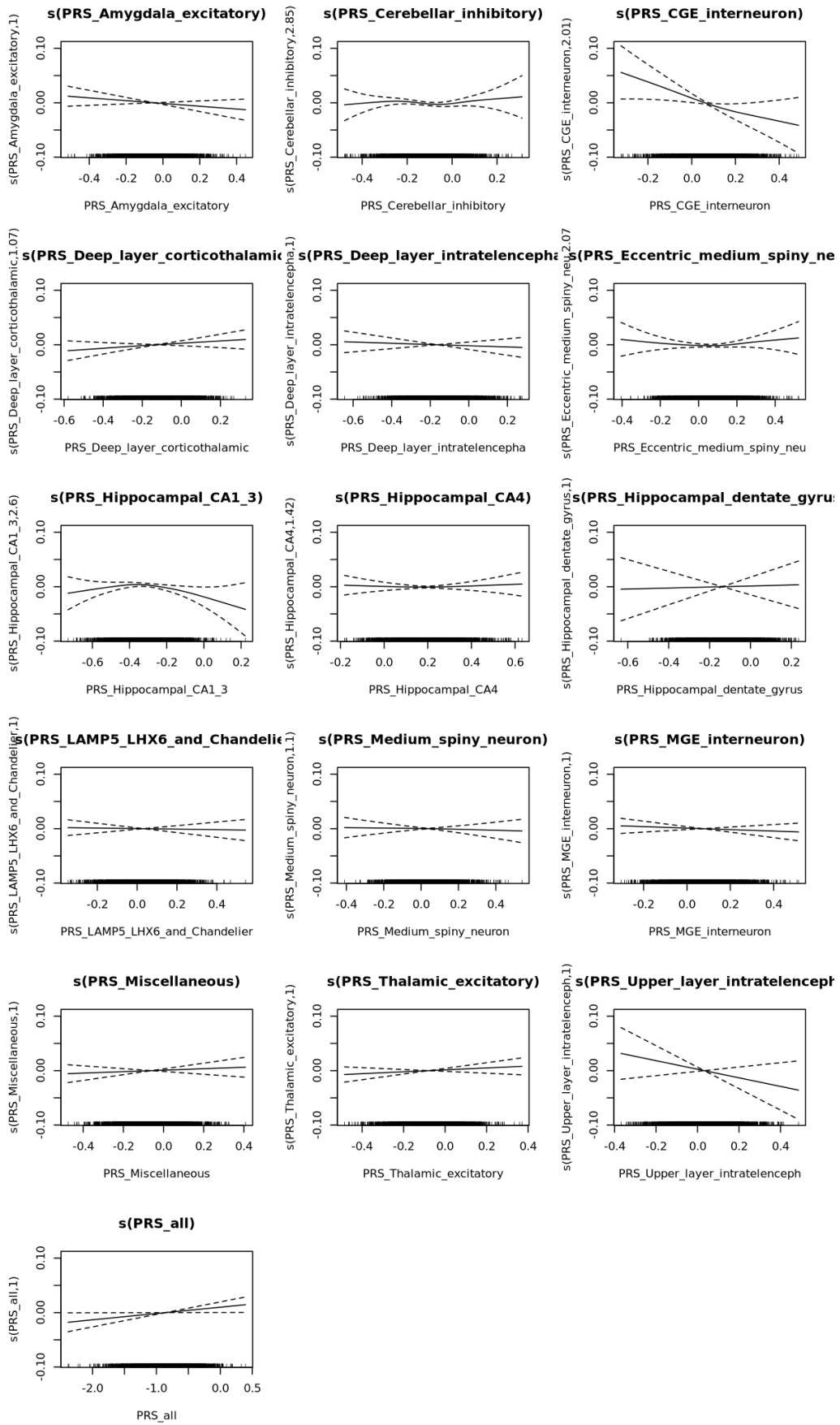


Figure C.3: Random effect of cell-type PRS smooth terms in Positive GAMM Model

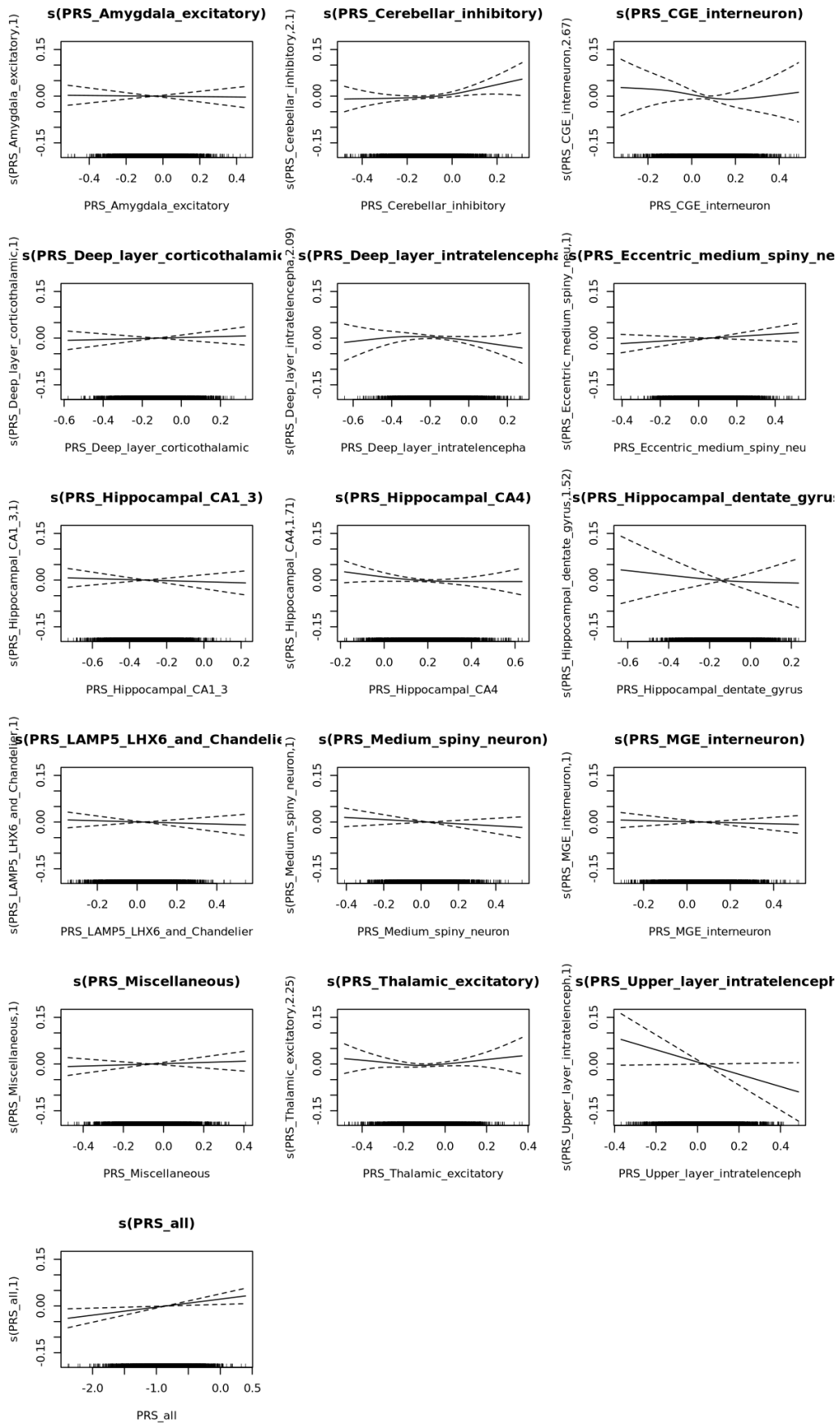


Figure C.4: Random effect of cell-type PRS smooth terms in Negative GAMM Model